



The geology of the Greenstone orogenic gold deposit, Geraldton, Ontario, Canada: Structural controls, mineralogy, geochemistry, and geochronology

Zsuzsanna Tóth^{a,b,*}, Bruno Lafrance^a, Benoît Dubé^c, Patrick Mercier-Langevin^c, Robert A. Creaser^d, Matthew I. Leybourne^{a,e}

^a Mineral Exploration Research Centre, Harquail School of Earth Sciences, Goodman School of Mines, Laurentian University, 935 Ramsey Lake Road, Sudbury, ON P3E 2C6, Canada

^b InnovExplo Inc., 560e Avenue, Val-d'Or, QC J9P 1S4, Canada

^c Geological Survey of Canada, Earth Sciences Sector, Natural Resources Canada, 490, rue de la Couronne, Québec, QC G1K 9A9, Canada

^d Department of Earth and Atmospheric Sciences, University of Alberta, 11223 Saskatchewan Drive NW, Edmonton, AB T6G 2E3, Canada

^e Queen's Facility for Isotope Research (QFIR), Department of Geological Sciences and Geological Engineering, Queen's University, 36 Union Street, Kingston, Ontario K7L 3N6, Canada

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ABSTRACT

The Greenstone orogenic gold deposit is located in the Beardmore-Geraldton belt (BGB) along the boundary between the granite-greenstone Wabigoon subprovince and the metasedimentary Quetico subprovince of the Archean Superior craton, Canada. The deposit is hosted by ca. 2700–2694 Ma turbiditic sandstone, banded iron formation, and ca. 2694 Ma feldspar-quartz porphyry, which underwent strong deformation within the 1 km-wide Bankfield-Tombill deformation zone along the southern margin of the BGB. The deformation zone includes folds and cleavage that formed during early D₁ thrust imbrication of the BGB, S-shaped folds and fabrics that formed during D₂ sinistral transpression, and Z-shaped folds, fabrics, and localized shear zones that formed during D₃ dextral reactivation of the deformation zone. Gold mineralization is associated with folded, early-D₁, quartz-carbonate veins (V₁) and with NE- to E-striking, *syn*-D₂, tourmaline-quartz veins (V₂) as well as quartz-carbonate veins (V₃). The V₁ and V₃ veins are surrounded by sericite-carbonate-pyrite ± albite-rutile alteration halos, and the V₂ veins are surrounded by carbonate-tourmaline-pyrite ± pyrrhotite-chalcopyrite alteration halos. Gold was deposited during fluid-rock sulfidation reactions that resulted in the formation of inclusion-poor pyrite with Ni-Co-As primary crystallographic zoning and inclusion-rich pyrite enriched in Au and other metals (Ag-As-Bi-Co-Ni-Pb-Sb-Te). Hydrothermal alteration associated with the deposition of the veins produced a broad, up to 250 m wide, sericite-carbonate alteration envelope, with S, Te, As, W, and Bi as the best pathfinder indicators to gold mineralization. Contrary to previous studies, which attributed the formation of gold deposits in the BGB to late-D₃, our results suggest that gold was emplaced during early-D₁ and D₂ and involved multiple hydrothermal fluid pulses during several deformation events, as suggested for other major Archean orogenic gold camps associated with major fault zones such as the Timmins and Kirkland Lake camps in the Abitibi subprovince of the Superior craton.

1. Introduction

The Greenstone gold deposit, Ontario, Canada, is an orogenic gold deposit associated with major crustal scale faults, porphyry intrusions, and linear belts of fluvial to shallow marine sedimentary rocks. These features are also common to the Timmins-Porcupine, Kirkland Lake, Malartic, and Val d'Or camps of the Abitibi greenstone belt (Robert et al., 2005; Dubé and Gosselin, 2007; Dubé and Mercier-Langevin,

2020). On a regional scale, gold deposits in these districts and other world-class districts, such as the Kalgoorlie and Laverton camps in the Yilgarn craton, share similar features. On the deposit scale, the deposits are characterized by shear-hosted quartz-carbonate veins, iron-rich host rocks (Fe-rich tholeiites and banded iron formation), intense hydrothermal alteration (carbonate, sericite, chlorite ± albite), and disseminated sulfide (pyrite, arsenopyrite, pyrrhotite) (Groves et al., 1998, 2003; Goldfarb et al., 2001, 2005; Robert et al., 2005; Phillips and

* Corresponding author.

E-mail address: zsus70701@gmail.com (Z. Tóth).

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Powell, 2010). Depletions in Na₂O and additions of CO₂, K₂O and S characterize the alteration envelopes surrounding the deposits, which are typically enriched in the following pathfinder metals: Ag, As, W, B, Mo, Sb, Te and less commonly Bi (Groves et al., 1998; Dubé and Goselin, 2007; Lafrance, 2015). Less commonly, albitization dominates instead of sericite alteration, and consequently the alteration halos are enriched in Na₂O and depleted in K₂O (e.g. Kerr-Addison deposit; Kishida and Kerrich, 1987; Groves et al., 1998).

Gold deposits in the Geraldton camp have regional and deposit-scale features similar to large orogenic gold deposits, and two of these deposits, the Hard Rock and MacLeod-Cockshutt deposits, collectively produced 1,635,485 oz of gold from 1938 to 1966 (Mason and White, 1986). In the 1980's and 1990's, the Greenstone deposit played a central role as a type example of epigenetic, now orogenic, gold deposit in the debate on syngenetic versus epigenetic origin of gold in banded iron formation-hosted gold deposits (Kerswill, 1993 and references therein). Since then, the two contiguous deposits have been acquired by Equinox Gold and renamed the Greenstone deposit. The latter is in an advanced stage of development as an open pit mine with production planned for the first half of 2024 (<https://www.equinoxgold.com>). It is expected to produce 5 million ounces of gold over an initial 14-year period (*ibid*). Access to large mechanically stripped outcrops and to exploration drill core prior to the development of the open pit provided an opportunity to reassess the structural setting and footprint of the Greenstone deposit. In this paper, we: (1) define its structural setting and the relative timing of gold mineralization; (2) describe its alteration envelope and associated sulfide minerals; (3) characterize its geochemical footprint; and (4) present an interpretation for the emplacement of gold mineralization and formation of the deposit. Our results suggest that the Greenstone deposit is *syn*-deformational as previously suggested but formed during earlier structural events and associated gold mineralization than previously recognized.

2. Geological setting

2.1. Regional geology

The Beardmore-Geraldton belt (BGB) is a transitional terrane along the boundary between the granite-greenstone Wabigoon subprovince and the metasedimentary Quetico subprovince of the Archean Superior craton (Fig. 1). The BGB extends for roughly 120 km from Lake Nipigon to the west to Long Lake to the east. The BGB comprises six alternating, shear-zone bounded, generally north-younging, metavolcanic and metasedimentary panels, which are similar to the rocks of the Wabigoon and Quetico subprovinces, respectively (Ayres, 1969; Devaney and Williams, 1989). The 2724.9 ± 1.1 Ma (Hart et al., 2002) pyroclastic and volcanic flow rocks of the northern metavolcanic unit (NVU) have felsic to intermediate composition and predominantly tholeiitic geochemical affinity (Carter, 1987; Kresz and Zayachivsky, 1991; Shanks, 1993; Tomlinson et al., 1996). The central metavolcanic unit (CVU) consists of 2724.9 ± 1.2 Ma (Hart et al., 2002), mafic to intermediate, pyroclastic rocks as well as massive, pillowed, and amygdaloidal flows with calc-alkaline or tholeiitic geochemical affinity (Carter, 1987; Kresz and Zayachivsky, 1991; Shanks, 1993; Tomlinson et al., 1996; Lavigne, 2009). The southern volcanic unit (SVU) comprises pillowed and massive, mafic to intermediate, flows with tholeiitic affinities (Carter, 1987; Williams, 1987; Kresz and Zayachivsky, 1991; Shanks, 1993; Tomlinson et al., 1996; Lavigne, 2009). The metavolcanic rocks of the NVU, CVU, and SVU formed in geodynamic environments with trace element attributes similar to modern back-arc, island arc, and oceanic crust settings respectively (Tomlinson et al., 1996).

The three panels of metasedimentary rocks, that are, the northern, central, and southern metasedimentary units (NSU, CSU, SSU), represent a single, southward-prograding, *syn*-tectonic clastic foreland wedge that was deposited between ca. 2700 Ma and 2694 ± 1.0 Ma from detritus derived from the erosion of the eastern Wabigoon subprovince

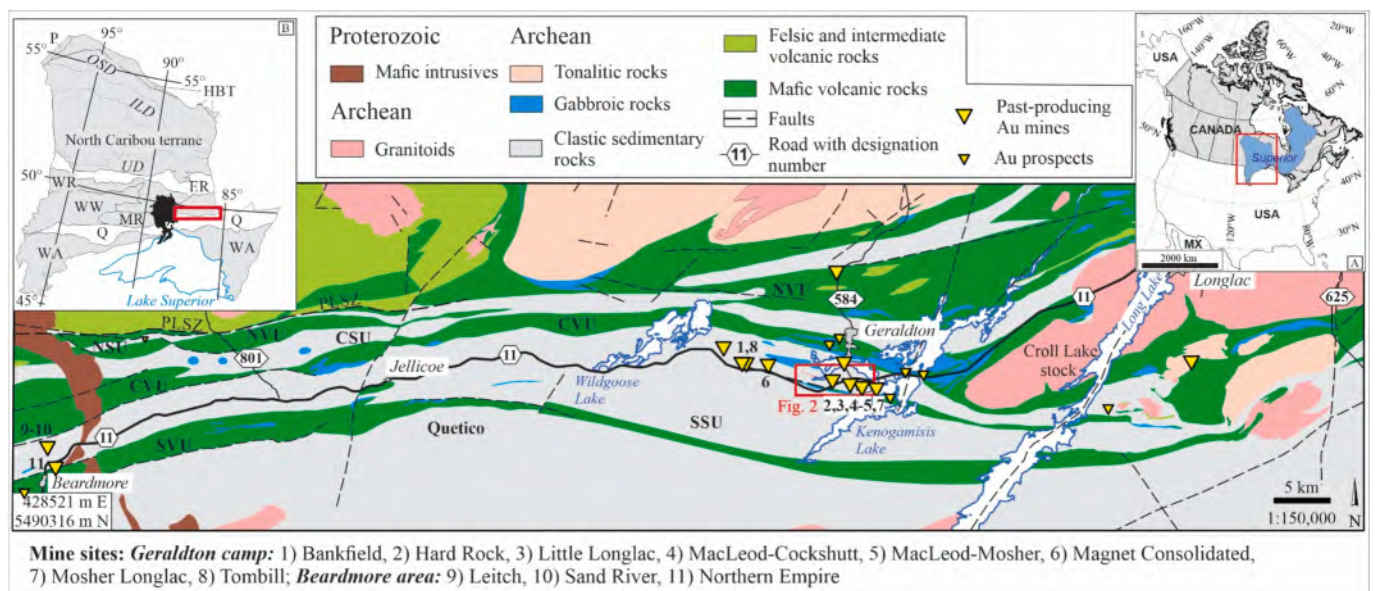


Fig. 1. Simplified geological map of the Beardmore-Geraldton belt (modified after Lavigne, 2009). PLSZ: Paint Lake Shear Zone. Abbreviations: NVU, CVU, SVU: northern, central and southern metavolcanic unit; NSU, CSU, SSU: northern, central and southern metasedimentary unit; MX: Mexico; USA: United States of America; Terranes: ER: English River; HBT: Hudson Bay; MR: Marmion; P: Pikwitonei; Q: Quetico; WA: Wawa-Abitibi; WR: Winnipeg River; WW: Western Wabigoon; Domains of North Caribou terrane: ILD: Island Lake Domain; OSD: Oxford-Stull Domain; UD: Uchi Domain. Map coordinates in UTM NAD 83, Zone 16. Inset A: the map of North America with the location of the Superior Province. The red rectangle shows the location of Inset B. (Montsion et al., 2018; https://www12.statcan.gc.ca/census-recensement/alternative_alternatif.cfm?l=eng&dispxt=zip&teng=lpr_000b21a_e.zip&k=%20%20%20130596&loc=/www12.statcan.gc.ca/census-recensement/2021/geo/sip-pis/boundary-limit/les-fichiers/lpr_000b21a_e.zip; <https://library.carleton.ca/find/gis/geospatial-data/shapefiles-canada-united-states-and-world/>; <https://www.natureearthdata.com/downloads/110m-cultural-vectors/110m-admin-0-countries/>) Inset B: the location of the BGB in the western Superior Province (after Percival et al., 2006). Lake Nipigon and Proterozoic rocks of the Lake Superior area in black. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(Devaney and Williams, 1989; Fralick and Pufahl, 2006; Tóth et al., 2022). The NSU consists of clast-supported fluvial conglomerate with minor sandstone (Mackasey, 1975, 1976; Devaney and Williams, 1989) unconformably deposited above the back-arc-like metavolcanic rocks of the NVU (Devaney and Fralick, 1985). The CSU comprises conglomerate, sandstone, interlayered mudstone, and sandstone with shallow water iron formation, representing subaqueous fan and/or prodelta environments (Pye, 1952; Horwood and Pye, 1955; Mackasey, 1975, 1976; Barrett and Fralick, 1985; Devaney and Williams, 1989; Fralick and Pufahl, 2006). The SSU is dominated by turbiditic mudstone, siltstone, and sandstone with minor iron formation and conglomerate lenses that were deposited in a submarine fan and/or basin plain environment (Pye, 1952; Horwood and Pye, 1955; Mackasey, 1975, 1976; Barrett and Fralick, 1985; Devaney and Williams, 1989).

Deformation of the BGB began as the southern Wawa-Abitibi greenstone terrane drifted northward and collided with the Wabigoon subprovince during the ca. 2690 Ma Shebandowanian orogeny (Devaney and Williams, 1989; Percival et al., 2012). This resulted in the closure of the Quetico oceanic basin and the accretion of the metavolcanic-metasedimentary panels of the BGB to the southern margin of the Wabigoon subprovince during early D₁ thrusting and folding. Dikes of the ca. 2694 Ma tonalite-trondhjemite-granodiorite (TTG) suite are *syn*-tectonic with the D₁ event, which ended with the emplacement of the stitching 2690 ± 1 Ma sanukitoid Croll Lake stock (Corfu, 2000; Lafrance et al., 2004; Tóth et al., 2022). This early accretionary event was followed by D₂ sinistral transpression and the formation of regional W-plunging F₂ folds, a regional W-plunging stretching lineation, an E-striking and steeply-dipping S₂ foliation, and D₂ high strain zones (Lafrance et al., 2004; Tóth et al., 2023a). Subsequent reactivation of the S₂ foliation and D₂ high strain zones during a late D₃ dextral transpression event resulted in the development of W-plunging Z-shaped F₃ folds, a W-SW-striking S₃ foliation, and dextral shear bands within localized dextral shear zones (Lafrance et al., 2004; Tóth et al., 2023a). The BGB is metamorphosed to greenschist facies, with metamorphism increasing to amphibolite grade along the margins

of intrusions (Kresz and Zayachivsky, 1991; Lavigne, 2009).

2.2. Deposit geology

Gold was discovered in the 1880's (Wilson, 1910) and the first mines, the Little Longlac mine in Geraldton and the Northern Empire mine in Beardmore, went into production in 1934 (Mason and McConnell, 1983; Mason and White, 1986). Over the following 30 to 40 years, the BGB produced over 4.1 Moz Au until mining ended in 1970 (Mason and McConnell, 1983; Mason and White, 1986). The host rocks, structural controls, and gold-silver production of the main mines are summarized in Table 1.

In the Geraldton camp (Fig. 2), most major past-producing gold mines are located in the Bankfield-Tombill deformation zone, which was interpreted as a major dextral fault (the Bankfield-Tombill fault; Fig. 2) along the boundary between the BGB and the Quetico metasedimentary subprovince (Pye, 1952; Buck, 1986; Williams, 1987; Lafrance et al., 2004). The Bankfield-Tombill deformation zone has been more recently reinterpreted as a dextrally reactivated D₂ high strain zone hosting late, parallel, dextral D₃ shear zones including the Bankfield-Tombill fault (Tóth et al., 2023a). Although gold deposits in the Geraldton area are known for their association with iron formation (e.g., Kerswill, 1996), significant production came from clastic metasedimentary rocks, felsic porphyritic and mafic intrusive igneous rocks (Table 1; Pye, 1952; Horwood and Pye, 1955). Gold mineralization occurs in E- to ENE-striking quartz-carbonate veins and their altered wallrocks along sheared contacts between metasedimentary rocks and intrusions, and in strongly folded quartz veins in F₃ fold hinges. As ore shoots plunge 20°–65° to the west, parallel to F₃ fold hinges, gold mineralization was previously attributed to the late D₃ dextral transpression event (Pye, 1952; Horwood and Pye, 1955; Anglin, 1987; Macdonald, 1988; Lafrance et al., 2004).

Table 1

Summary of the gold production in the BGB (summarized from; Pye, 1952; Horwood and Pye, 1955; Ferguson, 1965; Mason et al., 1985; Mason and White, 1986 and references therein).

map #	Mine	Operation years	Au (oz)	Ag (oz)	Host unit	Host rock	Mineralization
Geraldton area							
1	Bankfield	1937–1947	66,416	7,590	SSU	clastic metasedimentary, banded iron formation (BIF), felsic porphyry, diorite	East to NE-striking, steeply dipping quartz veins and altered wall rock; ore shoots plunge 65°W
2	Hard Rock	1938–1951	269,081	9,009	SSU	clastic metasedimentary, BIF, felsic porphyry, volcanoclastic rocks, peridotite, diorite	East to NE-striking, steeply dipping quartz veins and altered wall rock; ore shoots plunge 20–30°W
3	Little Longlac	1934–1956	605,449	52,750	SSU	clastic metasedimentary, BIF, diorite	Linear E- to ENE-striking ore zones which are locally deformed by S-shaped fold-like structures
4	MacLeod-Cockshutt	1938–1966	1,366,404	90,864	SSU	clastic metasedimentary, BIF, felsic porphyry, volcanoclastic rocks, diorite, gabbro	Confined to porphyry-greywacke contact; East to ENE-striking steeply-dipping quartz veins and the adjacent altered wall rock; parallel to axial plane of Z-shaped drag folds; ore shoot plunges at ca. 25–30°W
5	MacLeod-Mosher	1967–1970	180,576	17,321		Note: This mine was created from the amalgamation of the MacLeod and Mosher mines in 1967.	
6	Magnet Consolidated	1938–1952	152,089	16,879	SSU	clastic metasedimentary, BIF, felsic porphyry, diorite	Subvertical ENE-striking quartz veins and at one location ca. 45° W-plunging saddle-reef type veins
7	Mosher Longlac	1962–1966	330,042	34,586	SSU	clastic metasedimentary, BIF, felsic porphyry, diorite	East- to ENE-striking, intermediately to subvertically N-dipping quartz veins
8	Tombill	1938–1942	68,739	8,595	SSU	clastic metasedimentary, BIF, felsic porphyry, diorite	quartz stringers along altered and sheared contact between the porphyry and greywacke or BIF; east-striking steeply S-dipping shear hosted quartz veins within conglomerate
Beardmore area							
9	Leitch	1937–1965	861,982	31,802	SSU	clastic metasedimentary	Shear zone-hosted NE-ESE-striking moderately to steeply N-dipping quartz vein; ore shoots plunge 50°–65°W
10	Sand River	1937–1942	50,065	3,628		clastic metasedimentary	
11	Northern Empire	1934–1941	149,492	19,803	SVU	mafic to intermediate metavolcanic rocks	shear zone-hosted quartz vein striking ENE and dipping subvertically S
Total			4,100,335	292,827			

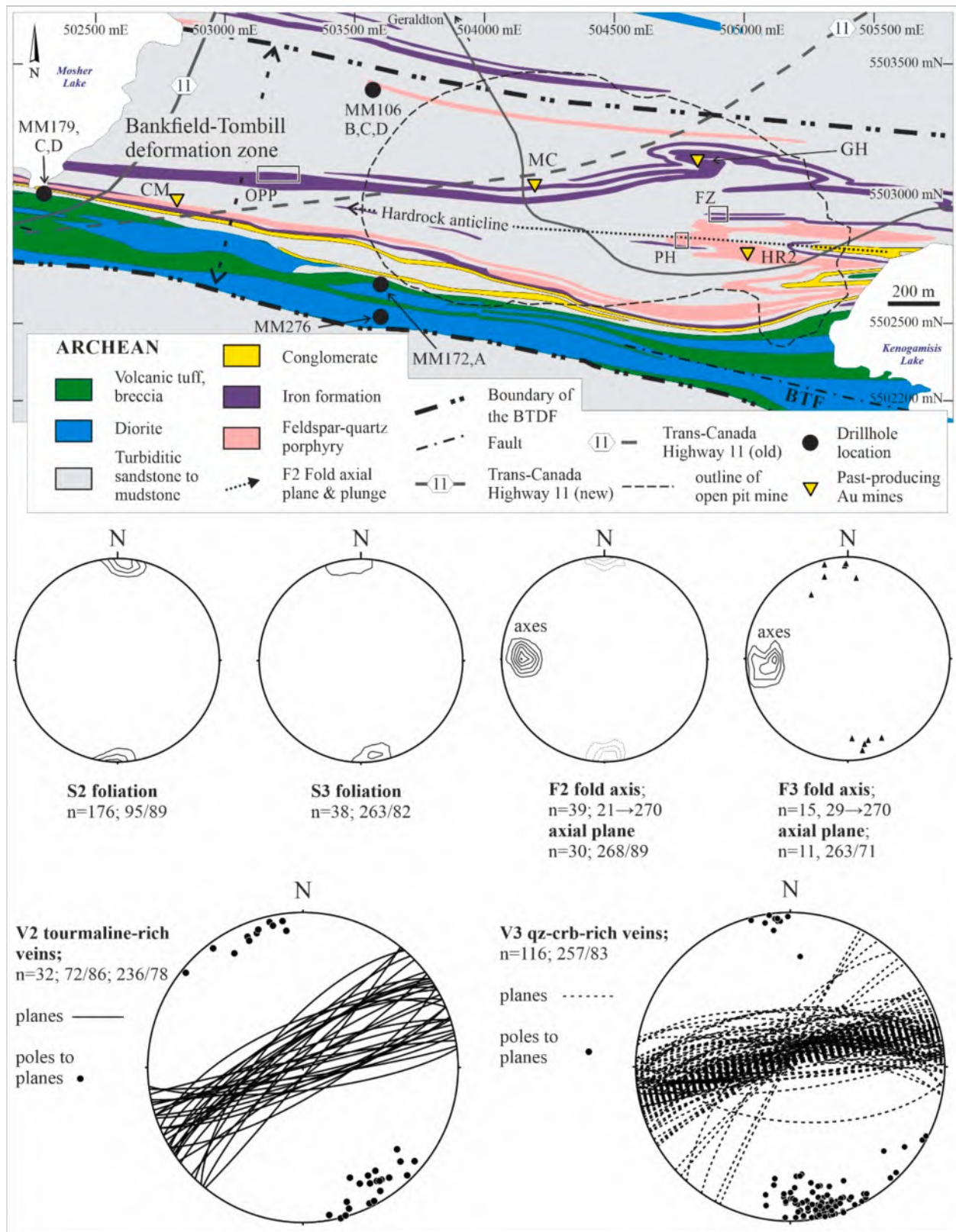
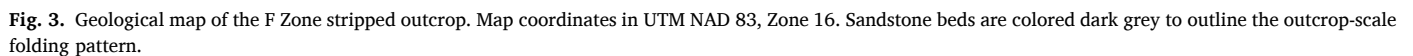


Fig. 2. Geological map of the Geraldton area (modified after [Pye, 1952](#); [Horwood and Pye, 1955](#)). The outline of the open pit is taken from <https://www.greenstonegoldmines.com>. The current trace of the Trans Canada Highway 11 is shown with solid line whereas the dashed line shows its historic location prior to the mine development. Outcrop Abbreviations: PH, Porphyry Hill; FZ, F Zone; OPP, OPP exposure. Past-producing mine sites: MC, MacLeod-Cockshutt; CM, Consolidated Moshier; GH, Glory Hole – Hard Rock open stope; HR2, Hard Rock shaft #2. Structures plotted on equal-area, lower hemisphere projections (n: number of measurements; average measurement to the right of the number of measurements; plotted using Stereonet software of [Allmendinger et al., 2013](#); [Cardozo and Allmendinger, 2013](#)). Contour intervals in 10% contour intervals per 1% area of the net.



3.1. Mineralogy and chronology of auriferous veins

Mineralization was exposed on three stripped outcrops on the Greenstone mine site. The stripped outcrops are located within the Bankfield-Tombill deformation zone (Fig. 2) and are informally named the F Zone (Fig. 3), OPP (Fig. 4), and Porphyry Hill (Fig. 5) outcrops. The hydrothermal gold-bearing system is characterized by early Fe-carbonate alteration and by three sets of auriferous veins, whose main features are summarized in Table 2. Early Fe-carbonate veins and alteration are present at the F Zone (Fig. 6A), OPP (Fig. 6B), and Porphyry Hill (Fig. 6C) outcrops where they are characterized by orange or rusty discoloration of sandstone beds and BIF and by the presence of massive Fe-carbonate veins crossed by quartz ladder veins. The Fe-carbonate veins and Fe-carbonatized beds are folded by F_1 folds at the F Zone (Fig. 6A) and are completely transposed parallel to the S_2 foliation on the OPP outcrop (Fig. 6B), where they are deformed into

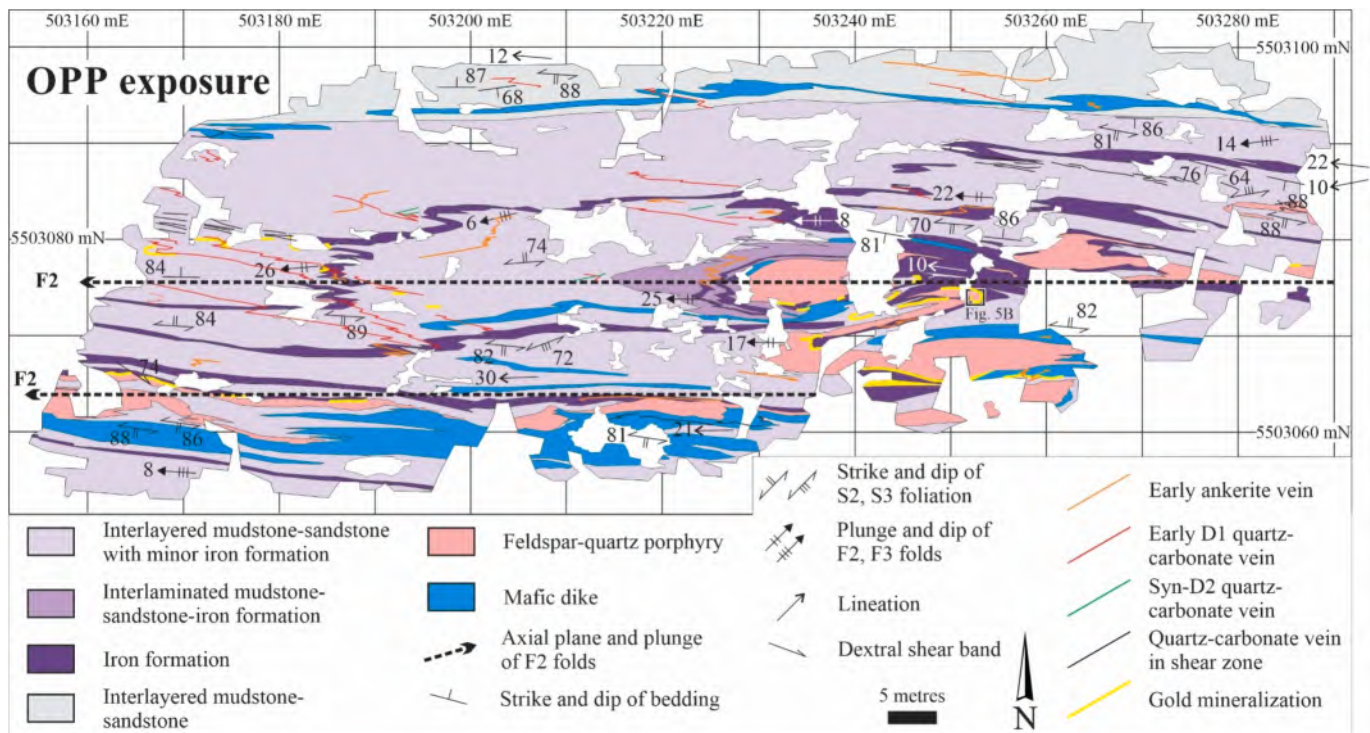


Fig. 4. Geological map of the OPP stripped outcrop. Map coordinates in UTM NAD 83, Zone 16.

discontinuous S₂-parallel lenses tapering to a wedge at both ends. This suggests that the veins and early carbonate alteration formed before or early during the D₁ deformation event.

An early set of bedding-parallel quartz-carbonate ($\pm$ pyrite, chalcopyrite, tourmaline) veins (V₁) is folded by F₁ and F₂ folds at the Porphyry Hill (Fig. 6D) and F zone (Fig. 6E) outcrops. At the OPP outcrop, the veins, which are outlined by solid red lines in Fig. 4, are oriented slightly clockwise to S₂ and bedding. They are deformed into boudins with narrow pinched boudin necks. The pinched boudin necks are folded within the hinge of F₂ folds (Fig. 6F), suggesting that the veins were emplaced and deformed prior to the D₂ event, likely during the D₁ event. V₁ veins are surrounded by strong alteration halos of sericite-carbonate-pyrite ($\pm$ rutile) in sandstone (Fig. 7A), sericite-pyrite-arsenopyrite ($\pm$ chlorite, biotite, carbonate) in BIF (Fig. 7B-C), and sericite-pyrite-carbonate-albite ($\pm$ rutile) in feldspar-quartz porphyry. The latter is pervasively altered to sericite, carbonate and disseminated pyrite, which increase in abundance towards the veins. In thin section, two textural types of pyrite are present: inclusion-rich pyrite rimmed by inclusion-poor pyrite (Fig. 7D). Chalcopyrite and native gold ($\pm$ ilmenite, rutile, tetrahedrite, galena, pyrrhotite, sphalerite, monazite, Au-Ag-tellurides) are present as inclusions predominantly in the inclusion-rich pyrite and fill fractures in both pyrite types. Rutile inclusions may contain finer-grained inclusions of Ag-Bi $\pm$ Pb-telluride and native gold (Fig. 7E), and inclusions of Au-Ag-tellurides are present in ilmenite grains within the matrix of feldspar-quartz porphyry. Arsenopyrite grains in BIF contain inclusions similar to those present in pyrite but have no native gold. Assays of V₁ veins yielded gold values ranging from nil in feldspar-quartz porphyry to a maximum of 65.1 g/t in BIF.

A second set of tourmaline-quartz veins (V₂) cuts across sandstone and feldspar-quartz porphyry. V₂ veins are steeply-dipping and NE- to ENE-trending (Figs. 2, 3). At the F Zone outcrop, V₂ veins cut across both limbs of F₂ folds (Fig. 8A). The veins overprint the F₂ folds and are in turn overprinted by gentle folds with the same S-shaped asymmetry as the F₂ folds (Fig. 8B). Other V₂ veins cut across the S₂ foliation on the south limb of the Hard Rock anticline exposed at Porphyry Hill (Fig. 8C). Adjacent to the veins, tourmaline has infiltrated the S₂ foliation and has

replaced minerals in sandstone and mudstone beds and the veins are overprinted by asymmetrical Z-shaped F₃ folds (Fig. 8D) that formed as flanking structures next to the veins, suggesting pre-D₃ emplacement of the veins. V₂ veins are surrounded by a carbonate-albite-pyrite alteration halo (Fig. 8E). In thin section, the veins locally contain molybdenite and monazite as well as pyrite, pyrrhotite, and chalcopyrite intimately intergrown with tourmaline. Inclusions of native gold are present in tourmaline and inclusion-poor pyrite (Fig. 8F, 9A).

A third set of steeply-dipping, NE- to E-trending, quartz-carbonate veins (V₃) overprint the hinge of F₂ folds (Figs. 3 and 9B). At the F zone, V₃ veins contain flakes of visible gold (Fig. 9C), cut across S-shaped F₂ fold hinges (Fig. 9B) and are folded by S-shaped folds (Fig. 9D). In banded iron formation at the OPP outcrop, V₃ veins and their strong sulfide-rich alteration halos strike 5° to 45° anticlockwise to bedding (Fig. 10A) and are oriented oblique to roughly parallel to the axial plane of later Z-shaped F₃ drag folds. Quartz-carbonate V₃ veins locally contain tourmaline, pyrite, chalcopyrite, xenotime, and monazite and are surrounded by alteration halo of sericite-carbonate-pyrite ($\pm$ arsenopyrite, rutile, ilmenite) in sandstone (Fig. 9C) and feldspar-quartz porphyry (Fig. 10B), and sericite-pyrite-arsenopyrite ($\pm$ carbonate, chlorite, biotite, monazite, pyrrhotite, chalcopyrite) in BIF (Fig. 10C). In thin section, pyrite is elongate parallel to the S₂ foliation and has inclusion-rich cores and inclusion-poor rims (Fig. 10D). Native gold occurs as inclusions in inclusion-rich pyrite (Fig. 10E), fills fractures and cavities in predominantly inclusion-poor pyrite and arsenopyrite (Fig. 10F), and fills fractures cutting through ferrodolomite along the margins of the quartz-carbonate veins. Chalcopyrite, pyrrhotite, ilmenite, and rutile ($\pm$ magnetite, galena, monazite, fluor-apatite, Au-Ag-, Ag- and Pb-Bi-tellurides, thorite) also occur as inclusions in pyrite (Fig. 10G) and arsenopyrite grains.

3.2. Geochemical footprint of the gold mineralization

The geochemical footprint of the mineralization is described for V₁ and V₃ veins at the F Zone and Porphyry Hill, where the structural relationship between the veins, fabrics, and fold generations can be most

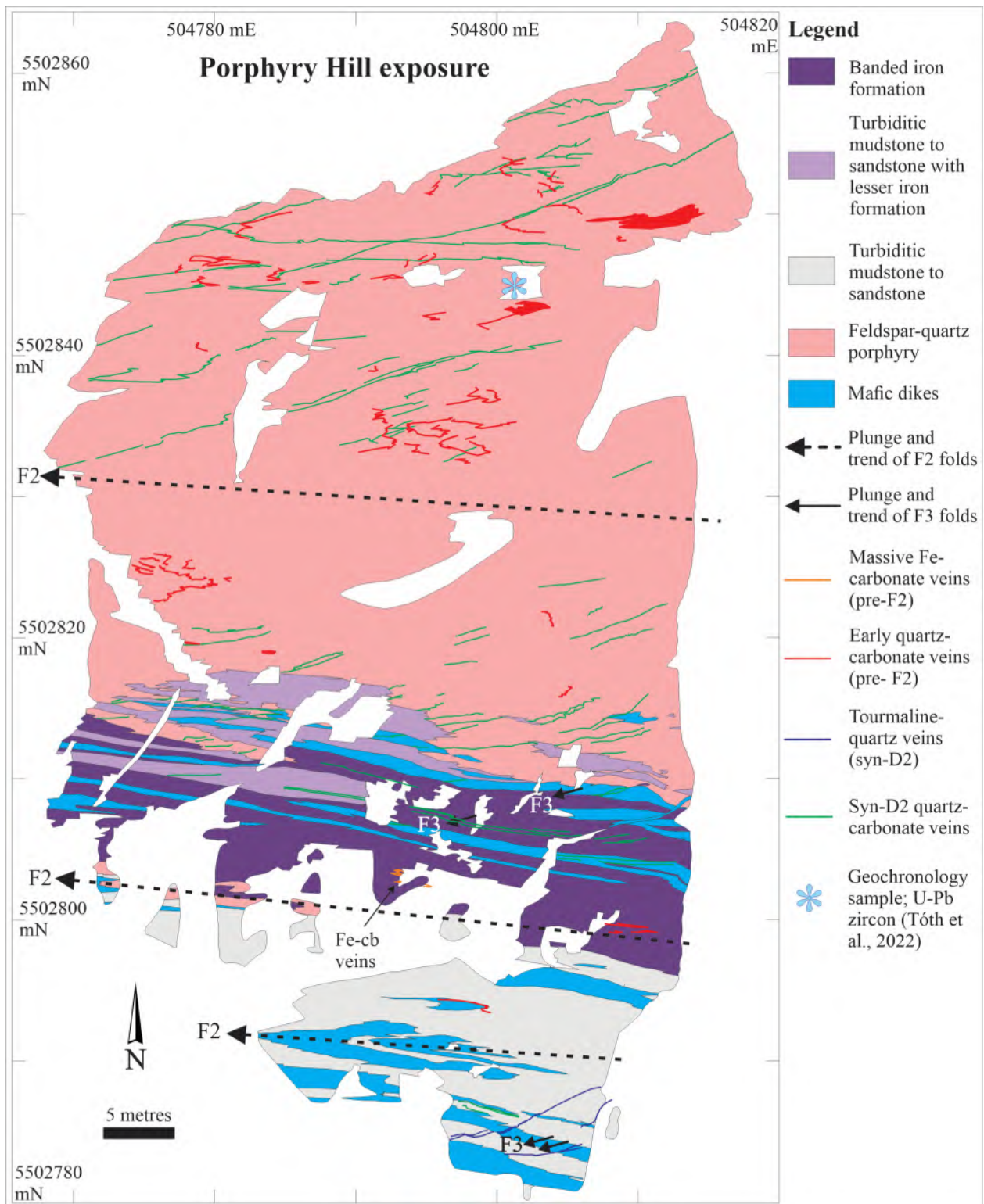


Fig. 5. Geological map of the Porphyry Hill stripped outcrop. Map coordinates in UTM NAD 83, Zone 16.

clearly demonstrated, and in two drill holes across high grade intervals in BIF and interlayered sandstone and mudstone.

One hundred and eighty-six drill core samples and 18 surface samples at the F Zone and Porphyry Hill were analyzed for major oxide and trace element compositions using X-ray fluorescence spectrometry (XRF) and inductively coupled plasma mass spectrometry (ICP-MS) methods, respectively. The analytical techniques and a summary of the

accuracy and precision tests are described in detail in [Appendix A](#). The precision and accuracy values were calculated based on analyses of reference materials and duplicates. The geochemical composition of representative samples is presented in [Supplementary Material 1](#).

Table 2
Characteristics of the gold mineralization events by host rock.

	Relative timing	Mudstone to sandstone-hosted	Banded iron formation-hosted	Porphyry-hosted
V ₁ auriferous quartz-carbonate-pyrite-chalcopyrite ± tourmaline vein	Early-D ₁	pervasive sericite-pyrite-quartz-rutile alteration halo; inclusions of chalcopyrite, ilmenite, rutile, tetrahedrite, galena, monazite in pyrite	Pyrite ± arsenopyrite-sericite ± chlorite ± biotite-carbonate alteration; pyrrhotite, native gold, chalcopyrite and magnetite inclusions in pyrite and arsenopyrite	quartz-albite-Ca-Mg-Fe-carbonate-pyrite-rutile alteration halo; inclusions of chalcopyrite, native gold (Au:Ag = 5–6:1), ilmenite, rutile, galena, sphalerite, and Ag-Bi-tellurides in pyrite
V ₂ auriferous tourmaline-quartz ± pyrite ± pyrrhotite ± chalcopyrite ± molybdenite ± monazite vein	Syn-D ₂	Ca-Mg-Fe-carbonate-tourmaline-pyrite-pyrrhotite-chalcopyrite, ilmenite, rutile and monazite alteration halo; Au-Ag-tellurides and galena inclusions and fracture-fillings in As-Pb-enriched pyrite;	no apparent alteration halo	albite-carbonate-pyrite-rutile alteration; rutile inclusions in pyrite
V ₃ auriferous quartz-carbonate-tourmaline-pyrite-chalcopyrite ± xenotime ± monazite vein	Syn-D ₂	strong sericite – carbonate – pyrite ± arsenopyrite alteration; inclusions of native gold, pyrrhotite, chalcopyrite, ilmenite, rutile, galena in pyrite	Pyrite (± pyrrhotite ± arsenopyrite)-quartz-sericite-chlorite-biotite-carbonate-monazite alteration; magnetite, chalcopyrite, pyrrhotite, and rarely ilmenite, rutile inclusions in pyrite	Ca-Mg-Fe-carbonate-albite-pyrite-rutile-ilmenite alteration; rutile, ilmenite, pyrrhotite, chalcopyrite, native gold, fluor-apatite, galena, monazite, Au-Ag-, Ag- and Pb-Bi-tellurides and thorite inclusions in pyrite

3.2.1. Geochemical footprint of mineralization at the F zone and porphyry Hill

Following Grant's (1986) isocon method, major and trace element compositions of altered samples were plotted against least-altered precursors to determine elemental changes in altered wallrocks of V₁ and V₃ veins emplaced in sandstone and feldspar-quartz porphyry. Elemental mobility and mass changes were assessed using generally immobile or least mobile elements, such as Zr and Hf in metasedimentary rocks and TiO₂ and Al₂O₃ in feldspar-quartz porphyry. This assessment suggests no to negligible mass changes associated with V₁ and V₃ vein mineralization in sandstone and feldspar-quartz porphyry. Elemental concentrations of altered samples and least-altered precursors were scaled and plotted on isocon diagrams and % elemental mass changes were corrected using the mass factor and plotted as histograms. The results are displayed in Figs. 11 and 12 and summarized in Table 3. Alteration halos

associated with V₁ and V₃ veins in sandstone and siltstone show major gains in Au, Ag, S, As and Sb (± Se, Te, Bi, CO₂). Alteration haloes associated with the same veins in feldspar-quartz porphyry show significant gains in Au, As, Te, Cu (± W, Se) and losses in Ag, In, Cd, Li (± Mo).

3.2.2. Diamond drill core sections across mineralized intervals

Spearman Rank correlation coefficients were calculated to establish element associations and enrichments in gold-mineralized zones. Correlations between gold and other elements from 186 drill core and 18 grab samples are presented in Table 4. Major oxide and trace element geochemical profiles are presented in Figs. 13 and 14 as well as in Supplementary Material 2. Positive correlations ($r > 0.5$) with 95 % confidence level ($p < 0.05$), which are listed in order of decreasing Spearman Rank correlation coefficients, are observed between Au and (1) S ($r = 0.9$), W, As, Bi, Te, ($r = 0.57$) in 78 sandstone samples; (2) Pb ($r = 0.79$), S, FeO, Te ($r = 0.60$) in 33 BIF samples; and (3) W ($r = 0.64$), As, Te, S ($r = 0.47$) in 30 porphyry samples. Collectively, Au grades have strong positive correlation with S ($r = 0.89$), W, As, Bi, and Te ($r = 0.59$) in all lithologies. Weaker positive correlations ($0.29 < r < 0.4$) are observed between Au and Pb, Sb, FeO, and Se.

Similar enrichments in Sb, W, Pb, Bi, Te, and As are observed in drill hole profiles through mineralized intervals and are plotted in Fig. 14 and Supplementary Material 2. The concentration of the pathfinder metals closely resembles the distribution of gold and remains slightly elevated for a few tens to 120 m beyond the mineralized zone. Hydrothermal alteration indices, which monitor the intensity and type of alteration, are also plotted and their Spearman Rank correlation coefficients with Au are presented in Table 4. Those providing the best vectors towards gold mineralization (at 95 % confidence level ($p < 0.05$)) are the sulfidation indices such as $S/(SiO_2 + Na_2O)$, $SSI = S/(S + Na_2O)$, and $KSI = S/(S + K)$, which emphasize the gains in total sulfur relative to other elements, and the sulfidation-carbonation index $(LOI + CO_2 + S)/SiO_2$ (Gemmell, 2006, 2007). At the deposit scale, the sulfidation and sulfidation-carbonation indices only show elevated values within the ore zone (≥ 0.3 g/t); therefore, they best delineate the contour of the ore body.

Table 4 shows that the sericitization ($MSI = 3 K/Al$ molar ratio, Kishida and Kerrich, 1987; $EPAI = K/(Al + Na)$ molar ratio, Whitbread and Moore, 2004), and carbonation (CO_2/Ca ; Eilu et al., 2001) indices show weak to moderate positive correlation with Au. This is consistent with diamond drill hole profiles (Fig. 14 and Supplementary Material 2) that outline an up to 250 m wide sericite and carbonate alteration envelope around the gold deposit. None of the alteration indices correlates strongly with Au in porphyry ($0.3 < r < 0.5$; Table 4), which is attributed to its strong to pervasive sericite-carbonate alteration irrespective of the presence of veins. Although the correlation between Au and carbonate or sericite alteration is weaker than that with sulfidation, these alteration indices can be used as vectors towards Au on district or deposit scale.

3.3. Laser-ablation ICP-MS trace element geochemistry of pyrite

The details of the analytical method are described in Appendix A. Trace element maps of pyrite grains from V₁ and V₃ veins in sandstone are shown in Figs. 15 and 16. They display consistent metal distributions between the pyrite grains from both vein generations. A pyrite grain associated with a V₁ vein is characterized by an inclusion-rich core and an inclusion-free idioblastic rim (Fig. 15). The latter exhibits a crude primary crystallographic zoning defined by variations in As, Co, and Ni concentrations, and these metals, together with Au, Ag, Bi, Cr, Cu, Pb, Sb, Te, Ti, V, and W, are enriched in the inclusion-rich core of the grain (Fig. 15). V₃ pyrite is similar to V₁ pyrite with an inclusion-rich core defined by enrichments in Au, As, Co, Ni, Pb, Sb, and Te (and to a lesser extent Ag and Bi) and an inclusion-free rim with a well-developed primary crystallographic zoning defined by variations in As, Co, Ni, and

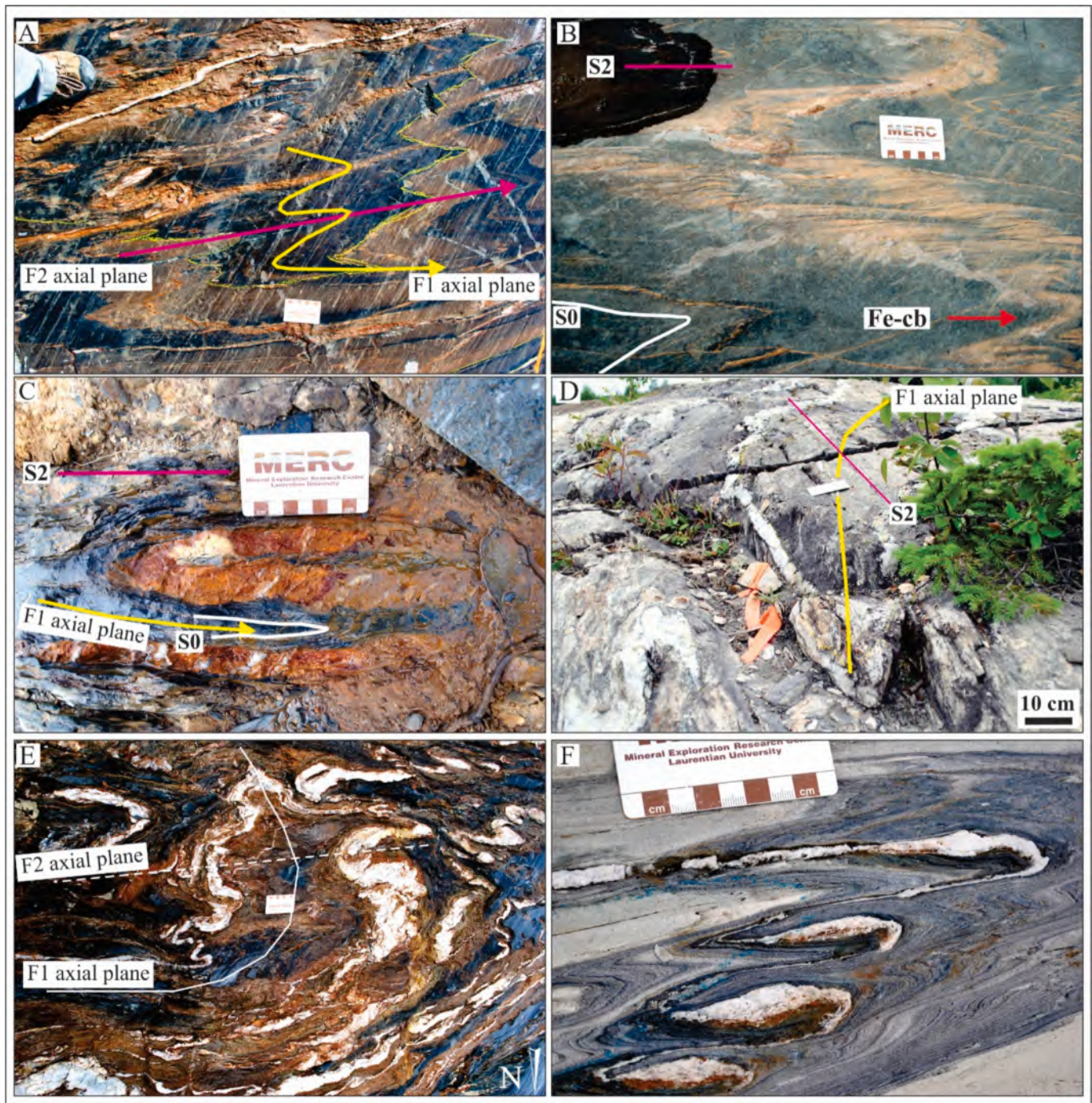


Fig. 6. Field photographs from Greenstone mine site. A) Bedding-parallel iron carbonate alteration folded by F_1 and F_2 folds in turbiditic sandstone at the F Zone outcrop; B) Bedding-parallel Fe-carbonate veins and alteration folded by F_2 fold and transposed parallel to S_2 cleavage at the OPP outcrop; C) Bedding-parallel Fe-carbonate vein folded by F_1 folds in BIF at the Porphyry Hill outcrop; D) V_1 vein folded by F_1 and F_2 folds in feldspar-quartz porphyry the at the Porphyry Hill outcrop. Photo card (9 cm in length) for scale; E) V_1 vein folded by F_1 and F_2 folds in sandstone at the F Zone outcrop. Photo card (9 cm in length) for scale; F) Boudinaged V_1 veins folded within F_2 fold hinge at the OPP outcrop. Photo card (9 cm in length) for scale. Abbreviations: S_0 : bedding; Fe-cb: iron carbonate.

possibly Au concentrations (Fig. 16). Both V_1 and V_3 pyrite grains show elevated Au-Ag-Bi-Te-Pb-Sb concentrations along fractures crosscutting the inclusion-poor pyrite or in inclusions within the generally inclusion-poor pyrite. In summary, inclusion-rich cores of V_1 and V_3 pyrite grains are typically enriched in Au, Ag, As, Bi, Co, Ni, Pb, Sb, and Te, whereas the inclusion-free rims of the grains have a well-defined to crude primary crystallographic Co, Ni, and As zoning.

3.4. Re-Os geochronology

A gold-mineralized, arsenopyrite-rich, feldspar-quartz porphyry sample (Bulk sample: 11.1 g/t Au, FQP without veins: 0.742 g/t Au; Tóth et al., 2023b; BGBZT2013.362C; Fig. 17A-B) with high Re content was chosen for arsenopyrite Re-Os dating. The feldspar-quartz porphyry sample consists of 35 % plagioclase and minor quartz phenocrysts (grain size: 0.5–2.5 mm; average: 1–2 mm) in a recrystallized, fine-grained matrix of quartz (58 %; average grain size: 30–50 μ m), carbonate

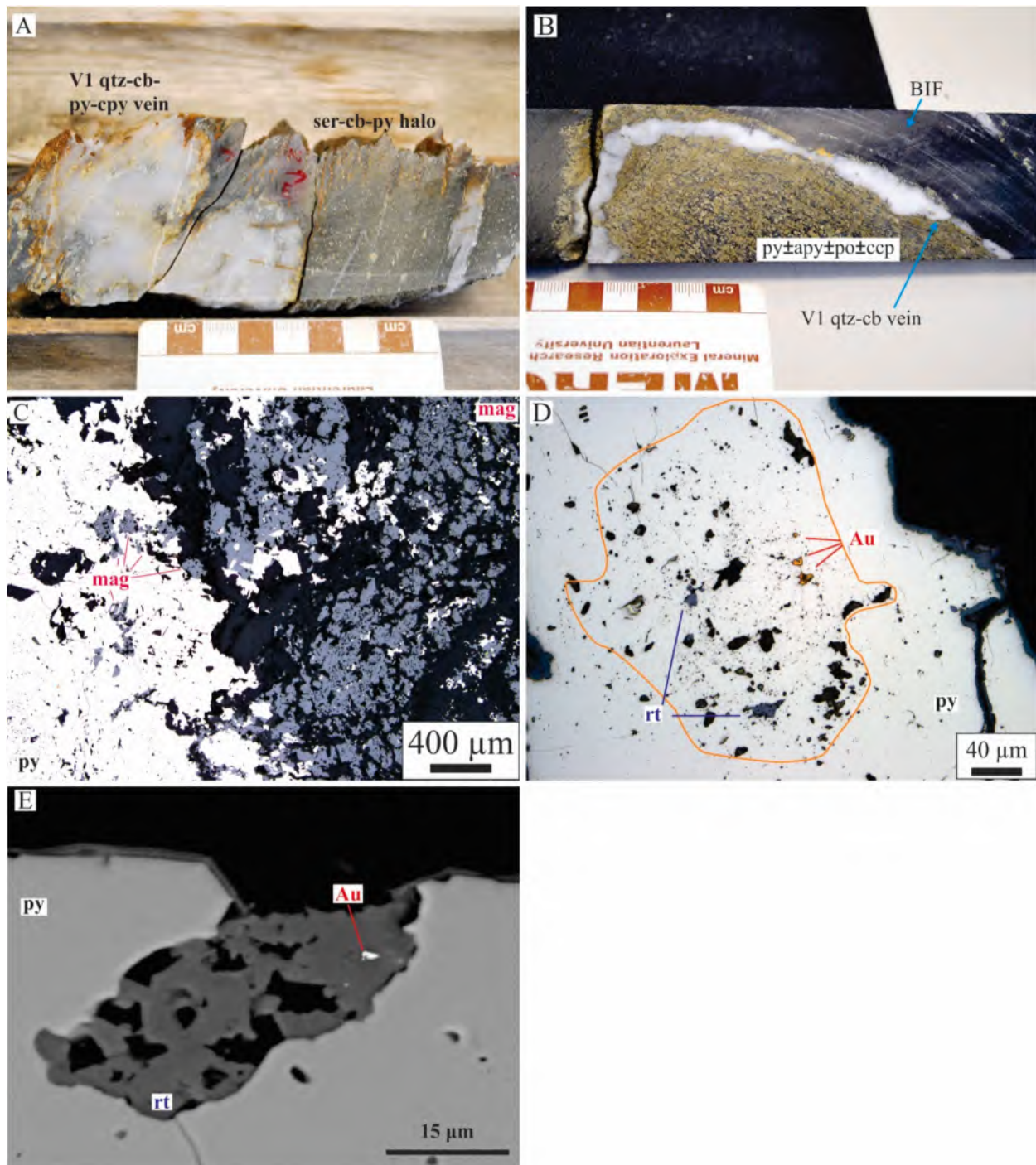


Fig. 7. Photographs of samples from the Greenstone mine site. A) Sericite-carbonate-pyrite alteration halo surrounding V₁ quartz-carbonate-pyrite-chalcopyrite ± tourmaline vein in turbiditic sandstone; B) V₁ vein surrounded by sulfide-rich replacement alteration halo in BIF; C) Reflected light photomicrograph of sulfide alteration halo shown in Fig. 7B, showing pyrite replacing magnetite. D) Reflected light photomicrograph of pyrite grain with an inclusion-rich (encircled in orange) core containing rutile (rt) and gold (Au) inclusions in feldspar-quartz porphyry host rock; E) Scanning electron microscope (SEM) backscattered image of rutile with native gold inclusion within pyrite from the wallrocks of V₁ veins in feldspar-quartz porphyry. Abbreviations: cb: carbonate; qtz: quartz; py: pyrite; ccp: chalcopyrite; ser: sericite; apy: arsenopyrite; po: pyrrhotite; mag: magnetite; BIF: banded iron formation; Au: native gold; rt: rutile (Kretz, 1983; Siivola and Schmid, 2007). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

minerals (7 %), and minor sericite, arsenopyrite, and pyrite (Fig. 17C). Arsenopyrite and pyrite occur as aggregates of multiple grains and as discrete, subhedral to euhedral, grains within quartz-carbonate veins along the contact between the porphyry and sandstone. They also occur as disseminations in both rock types (Fig. 17A-D). Six analyses were done (method described in Appendix A) and they define an isochron yielding a model age of 2579 ± 25 Ma (2σ) with an initial $^{187}\text{Os}/^{188}\text{Os} =$

0.58 ± 0.11 (MSWD = 0.63; Table 5; Fig. 17E).

4. Discussion

4.1. Structural controls of the gold mineralization

Previous studies on structural controls on gold mineralization in the

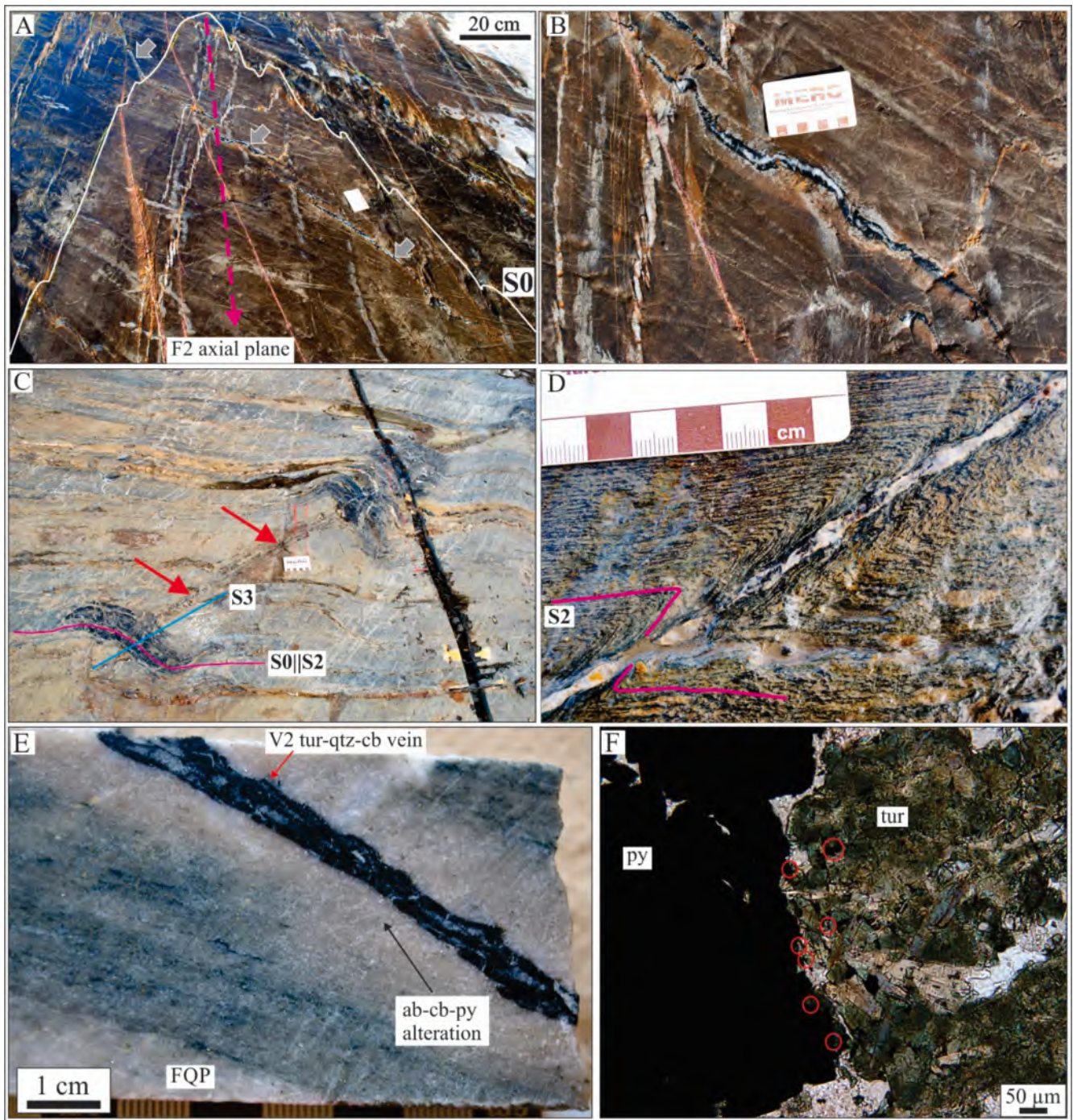


Fig. 8. A) Field photograph of V₂ vein cutting across the axial plane of F₂ fold in sandstone at the F Zone outcrop. The three grey arrows point to the vein. Photo scale (9 cm in length) for scale; B) Close-up of Fig. 8A. Note the small S-shaped folds overprinting the vein. Photo card (9 cm in length) for scale; C) Field photograph of tourmaline-quartz V₂ vein at the Porphyry Hill outcrop. Note replacement of wallrocks parallel to S₂ cleavage by tourmaline and the F₃ flanking fold along the margin of the vein. The red arrows point to the V₂ vein. Photo card (9 cm in length) for scale; D) Field photograph of Z-shaped F₃ flanking folds adjacent to tourmaline-rich V₂ vein in a mafic dike at the Porphyry Hill outcrop. Note the pinch-and-swell structures along vein. Photo card (9 cm in length) for scale; E) Drill core photograph of V₂ tourmaline-quartz vein surrounded by albite (ab)-carbonate (cb)-pyrite (py) alteration halo in feldspar-quartz porphyry (FQP). F) Plane polarized transmitted light photomicrograph of tourmaline-quartz V₂ vein with native gold inclusions (circled in red) in pyrite (py) and tourmaline (tur). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Geraldton area concluded that gold was introduced late during the structural evolution of the belt. Anglin (1987) described the presence of mineralized veins within sheared rock units, along sheared lithological contacts, and parallel to an earlier folded S₁ and a later S₂ cleavage (S₂ and S₃ in this study) and suggested that gold mineralization was introduced both prior to, and after, the formation of the S₂ cleavage and

postdated the emplacement of the porphyry dikes by 130 m.y. (Anglin, 1987).

Macdonald (1988) argued for the deposition of gold in conjugate Riedel fractures and P shear fractures during late dextral shearing. Lafrance et al. (2004) explained the plunge of the ore shoots parallel to F₃ fold axes by the emplacement of gold-mineralized veins in dextral

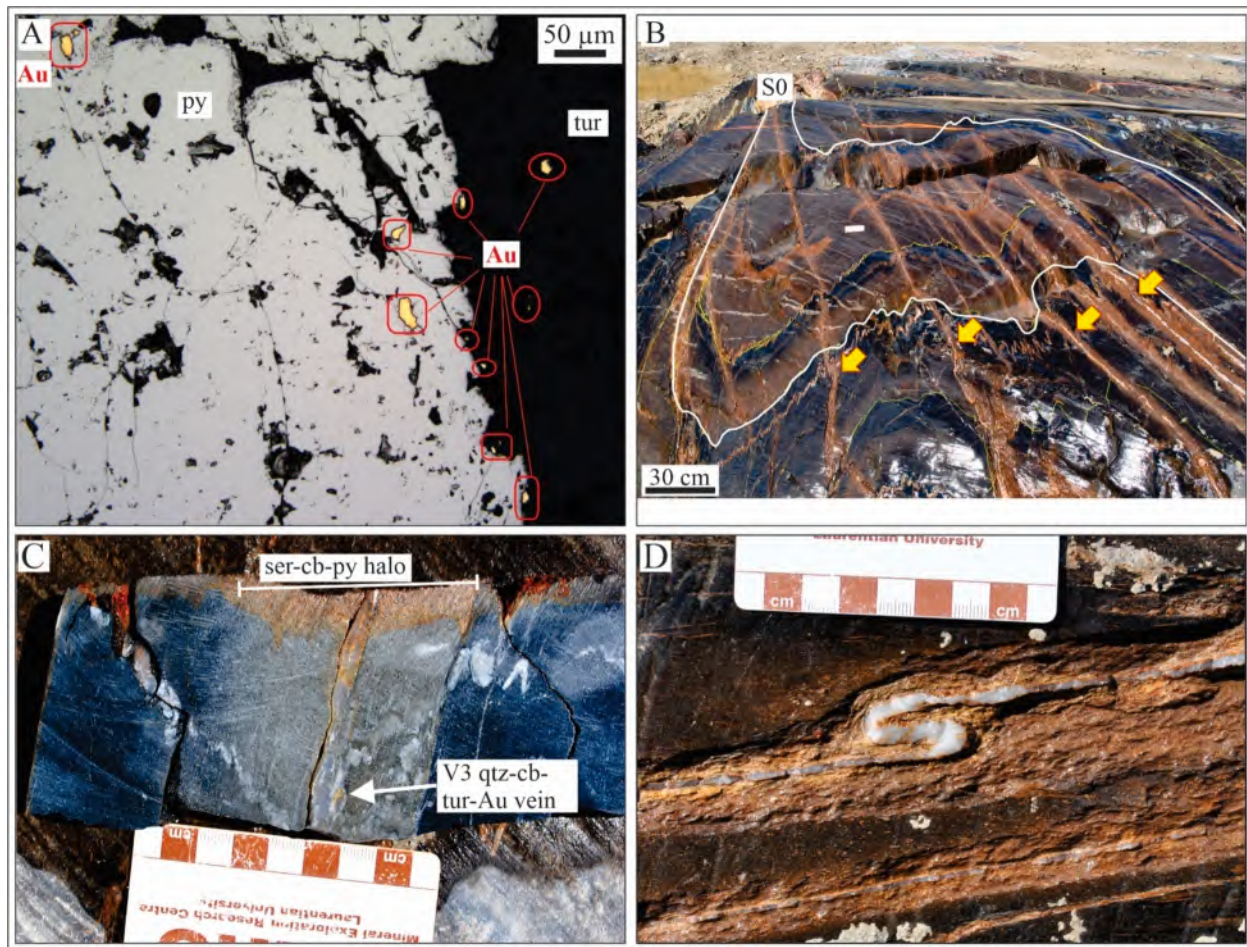


Fig. 9. A) Reflected light photomicrograph of tourmaline (tur) and pyrite (py) grains shown in Fig. 8F. Native gold (Au) inclusions within red circles and rectangles; B) Field photograph of quartz-carbonate V₃ veins (indicated by arrows) cutting across the axial plane of F₂ folds in sandstone at the F Zone outcrop. Photo card (9 cm in length) for scale; C) Photograph of hand sample showing a quartz-carbonate V₃ vein surrounded by sericite-carbonate-pyrite alteration halo in sandstone at the F Zone outcrop. Photo card (9 cm in length) for scale. D) Field photograph of quartz-carbonate V₃ vein overprinted by S-shaped F₂ fold at the F Zone outcrop. Photo card (9 cm in length) for scale. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

shear zones and along the axial planes of F₃ folds during D₃ dextral transpression. Lavigne (2009) described a gabbro xenolith with C-C' shear fabric in a gold mineralized porphyry dike and suggested that regional dextral shear zones acted as conduits for the emplacement of the dikes and later deposition of gold.

One of these shear zones, the Bankfield-Tombill fault is located along the southern margin of the Bankfield-Tombill deformation zone. Mechanical stripping of an outcrop exposed the fault as a late brittle-ductile structure hosting mineralization. Dextral shear bands and asymmetric strain shadows around volcanic fragments suggest that the fault formed during the D₃ deformation event (Tóth et al., 2023a). Sulfide minerals occupy Riedel shear fractures along the brittle-ductile fault, which is consistent with previous interpretations about the late deposition of at least some of the gold along brittle fractures (Macdonald, 1988) and within dextral ductile shear zones during the D₃ deformation event (Anglin, 1987; Macdonald, 1988; Williams, 1989; Lafrance et al., 2004).

Located 65 km west of Geraldton, the Brookbank gold deposit is another example of interpreted syn-D₃ mineralization. The deposit lies along a play of the Paint Lake deformation zone at the northern margin of the BGB. Gold is associated with quartz-carbonate veins, ankerite-sericite-chlorite-epidote-pyrite alteration zones, and quartz-orthoclase-pyrite alteration patches within dextrally sheared metabasalt (DeWolfe et al., 2007).

In the Geraldton area, recently stripped outcrops investigated as part of this study provide new information on the relative chronology and

structural controls on mineralization. Our work suggest that the bulk of the gold mineralization was introduced before the D₃ deformation event. Early Fe-carbonate altered beds and veins are folded by F₁ and F₂ folds, and where deformation was more intense, the beds were completely transposed parallel to the S₂ cleavage. As deformation facilitates fluid flow, fluid-rock interaction and alteration, the Fe-carbonate alteration likely occurred early during the D₁ event as strain became localized along the Bankfield-Tombill deformation zone during early thrust imbrication of the belt (Fig. 18A-C). Auriferous V₁ quartz-carbonate veins are also folded by F₁ and F₂ folds, and some veins were boudinaged prior to F₂ folding, suggesting that they were also emplaced early during D₁ thrusting (Fig. 18A-C).

Auriferous V₂ tourmaline-quartz and V₃ quartz-carbonate veins cut across the axial plane of F₂ folds as arrays of extensional veins oriented anticlockwise to bedding and to F₂ fold axial planes (Fig. 18D-F). Their orientation suggests that the veins were emplaced during NE-SW shortening, which is consistent with sinistral shearing along the ESE-striking Bankfield-Tombill deformation zone. As the veins cut across the S-shaped F₂ folds but are also S-folded, these observations collectively suggest that the veins were emplaced during sinistral shear along the Bankfield-Tombill deformation zone as the shear component of regional D₂ transpression became localized along the deformation zone (Tóth et al., 2023a). The older V₂ veins, which are oriented at a higher angle to bedding and to S₂ than the V₃ veins (Fig. 18F), were emplaced first, underwent anticlockwise rotation during sinistral shear, and were

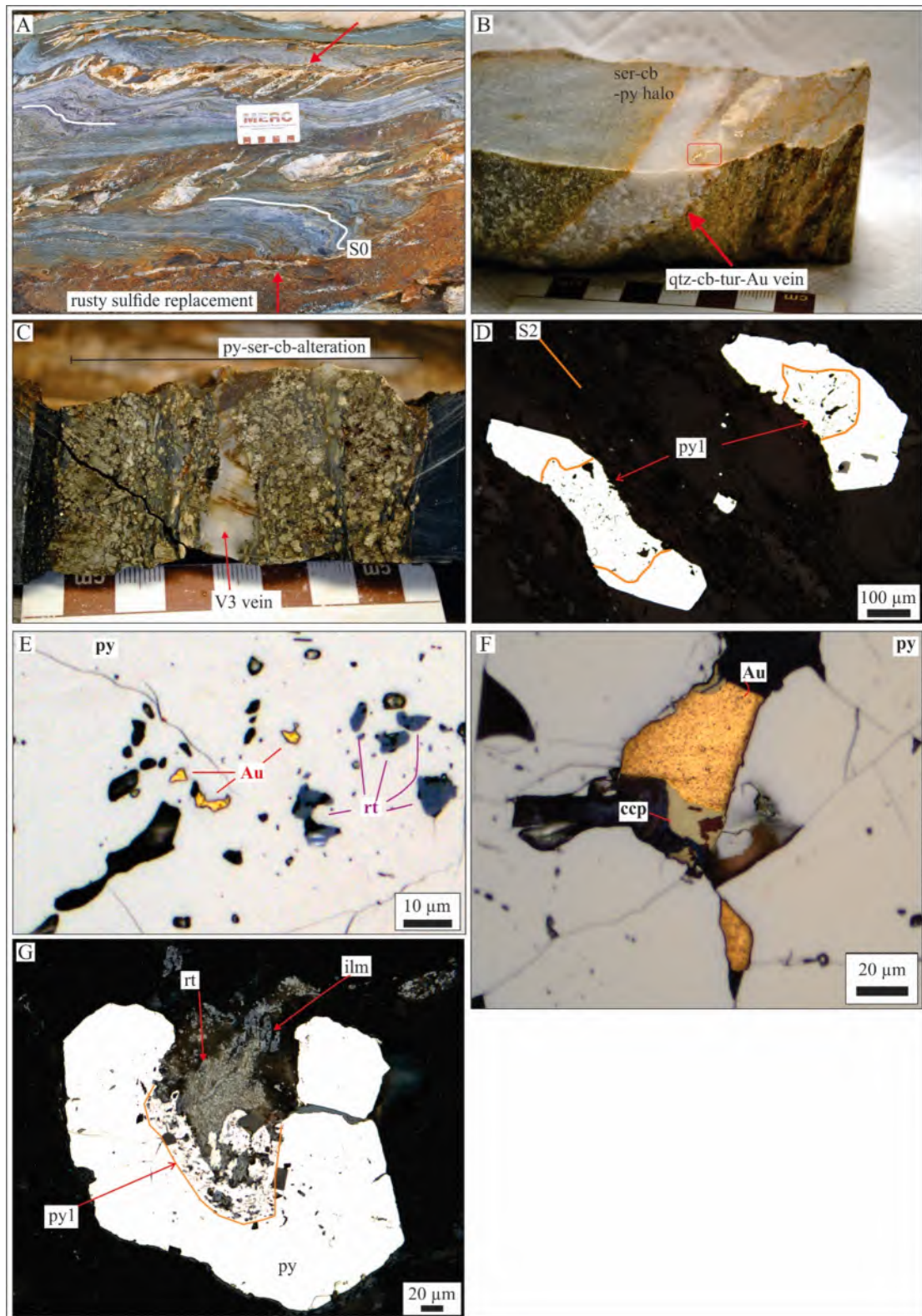


Fig. 10. A) Field photograph of en échelon quartz-carbonate V₃ veins (marked by red arrows) surrounded by sulfide-rich alteration halo in BIF at the OPP outcrop. Photo card (9 cm in length) for scale. B) Photograph of hand sample showing a quartz-carbonate V₃ vein with sericite (ser) – carbonate (cb) – pyrite (py) alteration halo in feldspar-quartz porphyry. Photo card (9 cm in length) for scale; C) Photograph of hand sample showing a quartz-carbonate V₃ vein with pyrite (py) – sericite (ser) – carbonate (cb) alteration halo in BIF. Photo card (9 cm in length) for scale; D) Reflected light photomicrograph of inclusion-poor pyrite surrounding an inclusion-rich pyrite core (pyl) in sandstone. Pyrite is elongate parallel to the S₂ foliation and is surrounded by pressure shadows filled with quartz; E) Reflected light photomicrograph of inclusion-rich pyrite (py) with native gold (Au) and rutile (rt) inclusions in py1 in sandstone wallrock of quartz-carbonate V₃ vein. F) Reflected light photomicrograph of visible gold and chalcopyrite filling fracture in inclusion-poor pyrite in BIF wallrock of quartz-carbonate V₃ vein; G) Reflected light photomicrograph of pyrite and rutile replacing ilmenite or titanomagnetite in quartz-feldspar porphyry wall rocks of quartz-carbonate V₃ vein. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

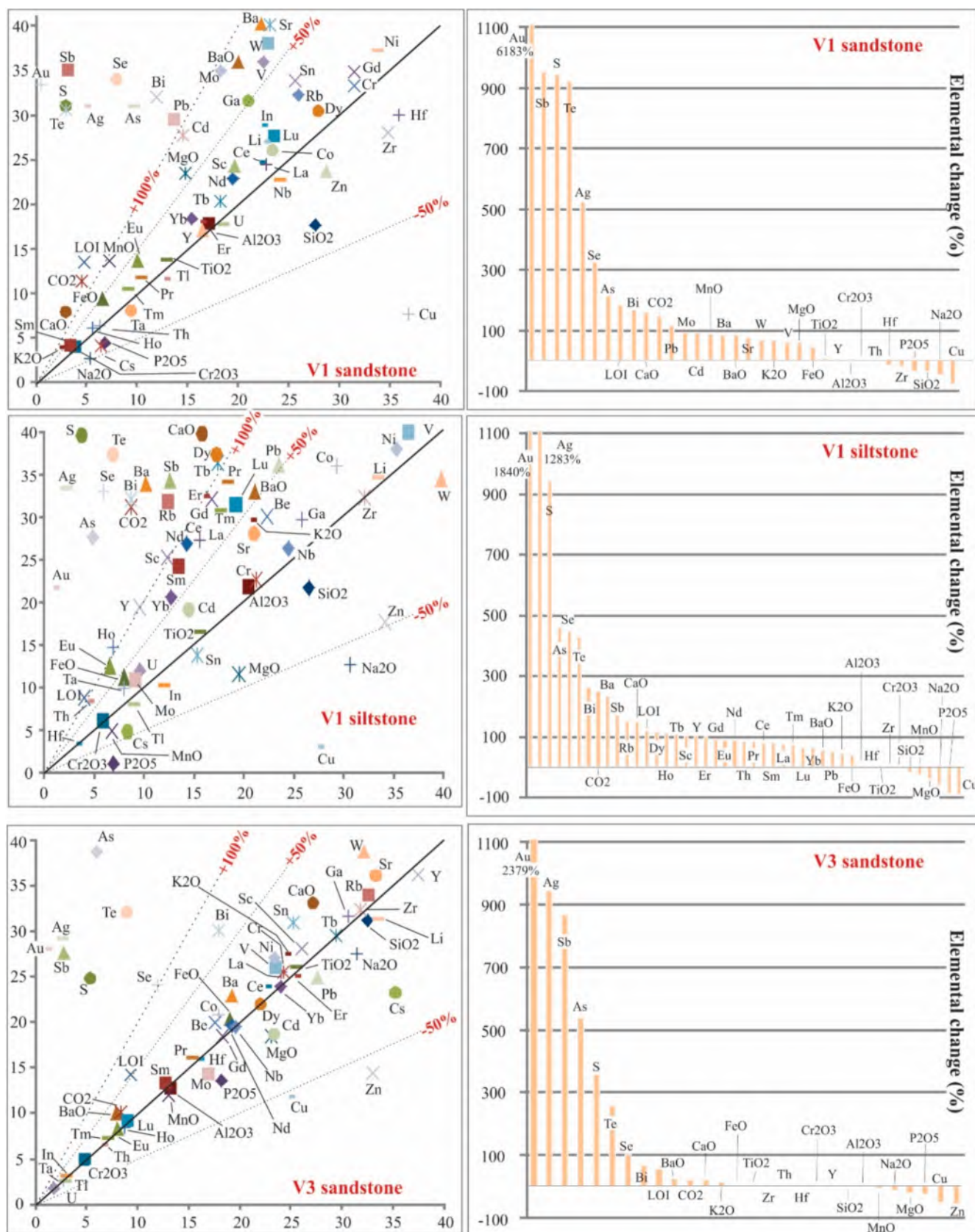


Fig. 11. Isocon diagrams and % mass change histograms showing mass losses and gains in sandstone and siltstone wall rocks of V₁ and V₃ quartz-carbonate veins. All major oxides, Zr, Th, Hf, Y and elements with % mass change greater than 50% are included in the histograms.

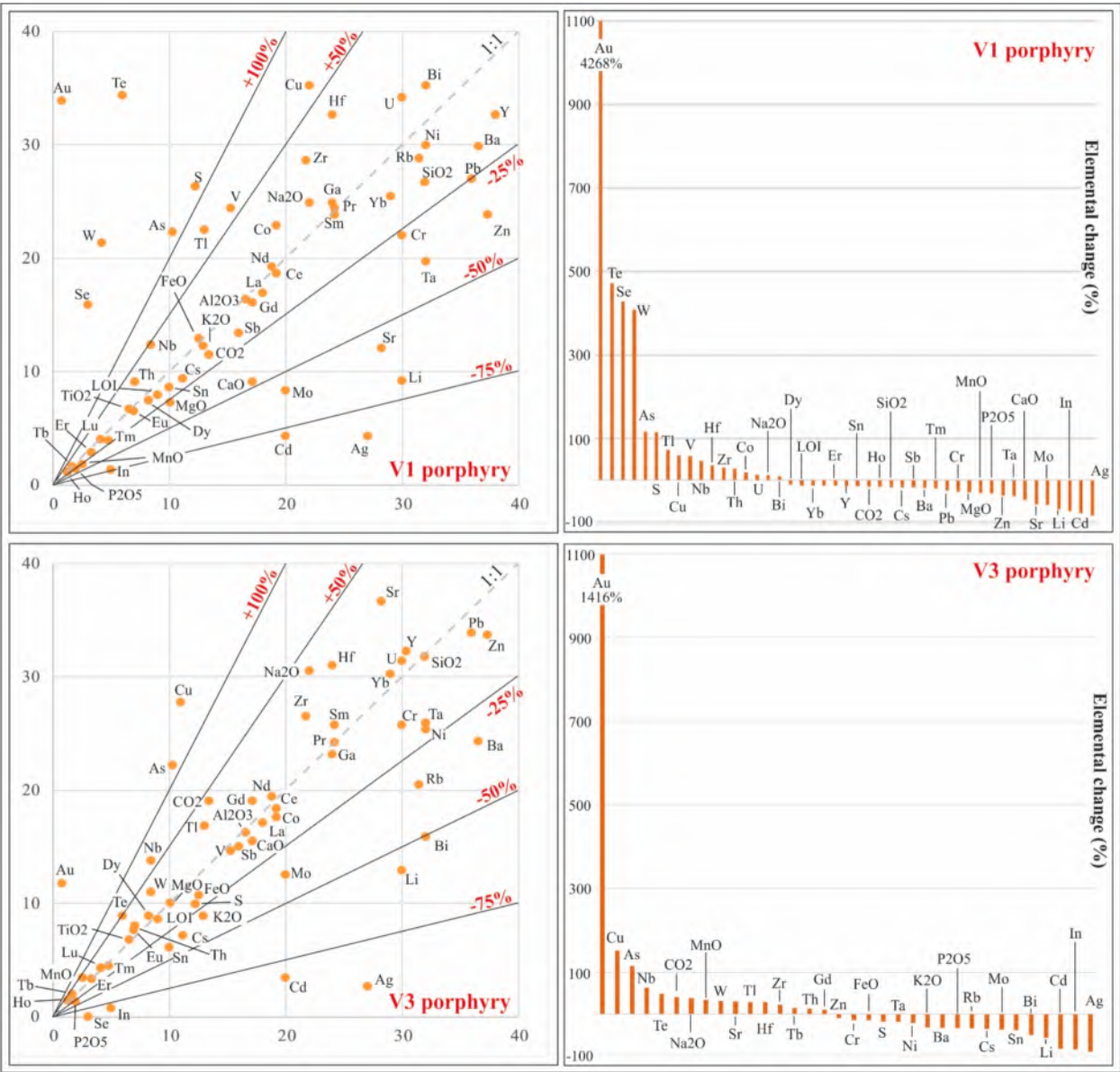


Fig. 12. Isocon diagrams and % mass change histograms showing mass losses and gains in feldspar-quartz porphyry wall rocks of V₁ and V₃ quartz-carbonate veins. All major oxides, Zr, Th, Hf, Y and elements with % mass change greater than 50% are included in the histograms.

Table 3
Elemental changes from less mineralized to more mineralized samples in sandstone, siltstone and porphyry during the early D₁ and syn-D₂ gold mineralization events based on Grant's isocon method (Grant, 1986). This list includes major oxides with at least 20 % and trace elements with at least 50 % elemental changes.

	Enrichment in:	Depletion in:
V ₁ in siltstone	Au, Ag, S, As, Se, Te, Bi, CO ₂ , Sb, Rb, CaO, LOI, Dy, Ho, Tb, Sc, Y, Er, Gd, Eu, Nd, Th, Pr, Ce, Sm, La, Tm, Le, Yb, BaO, Pb, K ₂ O, FeO	Na ₂ O, P ₂ O ₅ , Cu, MgO, MnO
V ₁ in sandstone	Au, Sb, S, Te, Ag, Se, As, LOI, Bi, CaO, CO ₂ , Pb, Mo, Cd, MnO, BaO, Sr, W, K ₂ O, V, MgO, FeO	Na ₂ O, Cu, SiO ₂ , P ₂ O ₅
V ₃ in sandstone	Au, S, Sb, As, S, Te, Se, Bi, LOI, BaO, CO ₂ , CaO	Cu, Zn, P ₂ O ₅ , MgO
V ₁ in porphyry	Au, Te, Se, W, As, S, Ti, Cu, V	Ag, Cd, In, Li, Mo, Sr, CaO, P ₂ O ₅ , MnO, MgO
V ₃ in porphyry	Au, Cu, As, Nb, CO ₂ , Na ₂ O, MnO	Bi, Li, Cd, In, Ag, K ₂ O, P ₂ O ₅

then overprinted by the V₃ veins which were emplaced later during the same D₂ shearing event. Subsequent dextral shear during the D₃ event reversed the rotation of the veins and produced flanking structures adjacent to the veins. The latter acted as planar anisotropies that impeded shearing along the S₂ cleavage and bedding and facilitated buckling and the formation of F₃ flanking folds (Fig. 18G). F₃ flanking structures adjacent to arrays of V₃ extensional veins are also present at Missing Link along the Paint Lake fault and in shear zones at the 801 stripping near Jellicoe (Tóth et al., 2023a).

The ore zones have shallow westerly plunges similar to those of other linear structural elements (stretching lineation, F₁, F₂ and F₃ fold axes) in the Bankfield-Tombill deformation zone. These structures and the ore zones formed before or during the dextral D₃ event and were rotated towards the shear direction during D₃ shearing (Tóth et al., 2023a). Thus, although the main ore zones formed early during the D₁ and D₂ events, their geometry reflects the total strain history of the Bankfield-Tombill deformation zone.

Table 4

Spearman Rank correlation coefficients between Au and various pathfinder elements and alteration indices in all samples combined and for each lithology.

Spearman correlation with Au	All samples (n=207)	Siltstone (n=41)	Sandstone (n=78)	BIF (n=13)	Sulphide-replaced rock (n=20)	BIF+sulfide replaced rock combined (n=33)	Porphyry (n=33)	Mafic dike (n=22)
FeO (wt. %)	0.29	0.33	-0.003	0.56	0.2	0.71	0.26	-0.35
CaO (wt. %)	-0.12	0.2	0.23	-0.54	0.14	-0.16	-0.029	0.8
K ₂ O (wt. %)	0.09	0.12	0.37	0.36	-0.072	0.076	0.017	0.56
LOI (wt. %)	0.35	0.24	0.33	0.3	-0.053	0.66	0.24	0.47
CO ₂ (wt. %)	-0.05	0.22	0.2	-0.42	-0.051	-0.13	0.28	0.55
S (wt. %)	0.89	0.9	0.9	0.81	0.15	0.75	0.47	0.66
Ba (ppm)	-0.018	0.028	0.17	0.23	0.024	-0.2	-0.27	0.6
Be (ppm)	0.11	0.12	0.5	-0.13	-0.31	-0.57	-0.16	0.63
Sb (ppm)	0.36	0.2	0.41	-0.025	-0.37	0.26	0.22	0.17
W (ppm)	0.73	0.71	0.74	0.6	-0.03	0.035	0.64	0.75
Cd (ppm)	0.15	0.092	0.24	NaN	0.46	0.39	0.29	0.45
Mo (ppm)	0.29	0.52	0.46	0.1	-0.065	-0.012	0.02	0.075
Pb (ppm)	0.38	0.065	0.24	0.49	0.44	0.79	-0.16	0.4
Bi (ppm)	0.6	0.49	0.62	0.66	-0.017	0.65	0.044	0.49
Te (ppm)	0.59	0.55	0.57	0.71	0.13	0.6	0.53	0.38
Se (ppm)	0.26	0.36	0.27	0.79	-0.068	0.15	0.089	0.41
Ag (ppm)	0.14	-0.18	0.053	0.27	0.39	0.63	0.032	-0.095
As (ppm)	0.61	0.33	0.66	0.56	-0.11	0.62	0.54	0.65
Zn (ppm)	-0.43	-0.39	-0.38	0.35	-0.26	-0.44	-0.15	0.072
Cu (ppm)	-0.17	-0.16	-0.22	0.25	-0.062	0.37	-0.062	-0.15
CO ₂ /Ca	0.24	0.33	0.27	0.03	-0.16	0.18	0.41	-0.0074
SSI: sodium-sulphide index	0.81	0.83	0.84	0.71	-0.13	0.67	0.4	0.75
MSI=3K/Al: muscovite saturation i.	0.43	0.36	0.38	0.32	0.026	0.11	0.0071	0.66
EPAl: potassic alteration i.	0.41	0.36	0.64	0.49	0.077	0.032	0.027	0.56
KSI: potassium sulphide i.	0.76	0.81	0.74	0.64	0.15	0.75	0.49	0.55
KVI: potassic volatile index:	0.13	0.086	-0.038	-0.17	0.11	0.52	0.22	-0.49
S/(SiO ₂ +Na ₂ O) mol	0.88	0.89	0.88	0.82	0.17	0.76	0.47	0.69
(LOI+CO ₂ +S)/SiO ₂	0.46	0.5	0.42	0.45	0.17	0.73	0.33	0.64

SSI: sodium-sulfide index: S/S + Na₂O molar ratio: $100 \cdot [S/32.066] / [(S/32.066) + 2 \cdot Na_2O/(61.978)]$; Gemmell et al., 2006, 2007.MSI: muscovite saturation index: $3 K/Al$ molar ratio: $[3 \cdot 2 \cdot K_2O/(94.196)] / [2 \cdot Al_2O_3/(101.961276)]$; Kishida and Kerrich et al., 1987.EPAl: potassic alteration index: $K/Al + Na$ molar ratio: $[2 \cdot K_2O/(94.196)] / [2 \cdot Al_2O_3/(101.961276) - 2 \cdot Na_2O/(61.978)]$; Whitbread and Moore, 2004.KSI: potassium sulfide index: S/S + K Molar Ratio: $100 \cdot [S/32.066] / [(S/32.066) + 2 \cdot K_2O/(94.196)]$; Gemmell et al., 2006, 2007.KVI: potassic volatile index: $100 \cdot LOI/(LOI + K_2O)$ molar ratio: $100 \cdot [LOI/18.01528] / [(LOI/18.01528) + 2 \cdot K_2O/(94.196)]$; Gemmell et al., 2006, 2007.

4.2. Hydrothermal alteration and geochemical footprint at the Greenstone gold deposit

Similar sericite-carbonate-pyrite alteration halos are associated with quartz-carbonate V₁ and V₃ veins although they were emplaced during

separate deformation events and in different rock types, i.e. sandstone and feldspar-quartz porphyry. The presence of pyrite in the vein alteration halos explains the mass gains in S and Se; the latter substitutes for S in sulfide minerals. Mass gains in CO₂ and CaO in sandstone are represented by the formation of carbonate minerals and mass gains in K₂O is

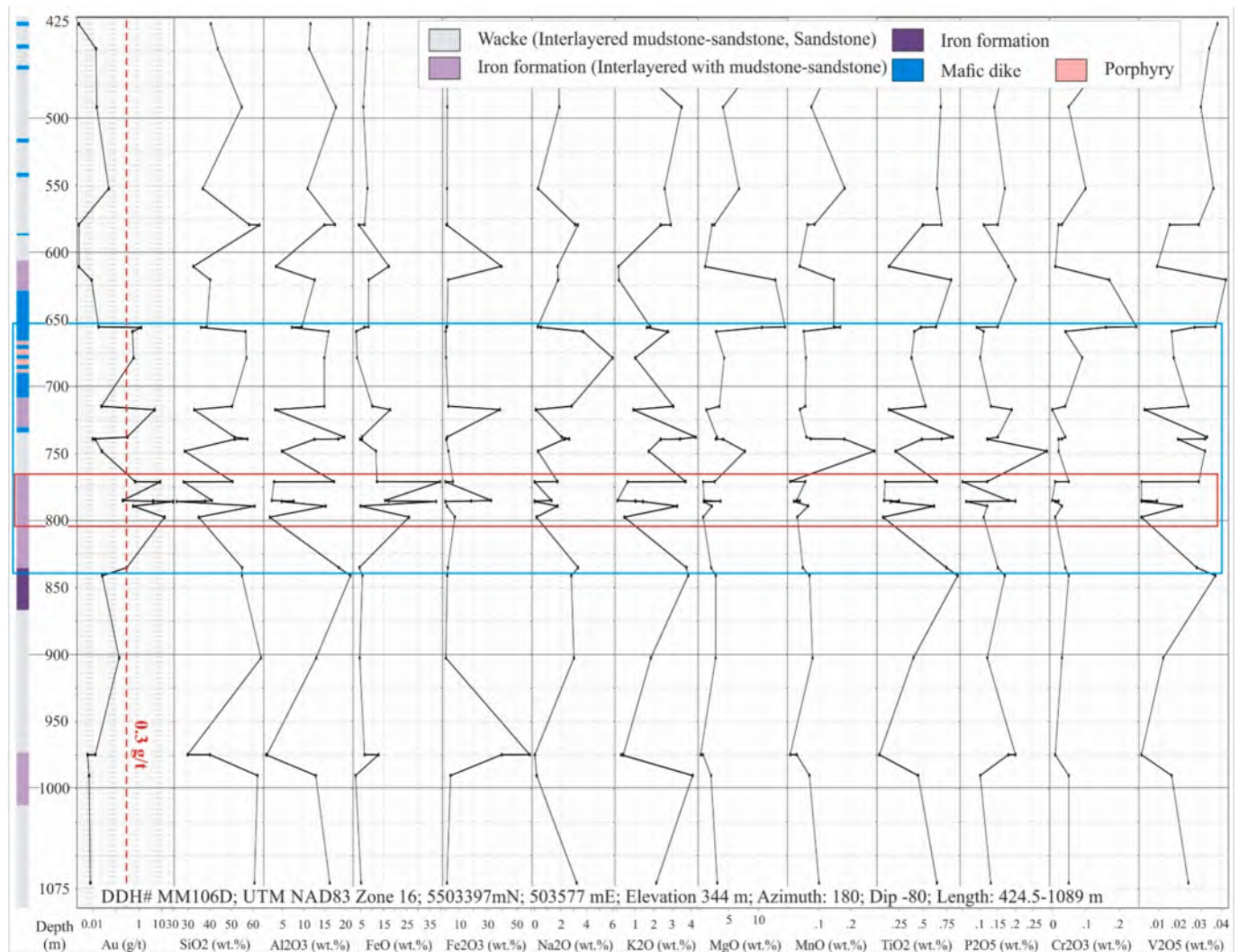


Fig. 13. Geochemical profile of major element oxides in drill hole # MM106D across section 503575E, UTM NAD 83, Zone 16. Fig. 2 shows the location of the drill hole. The blue box outlines the interval where Au exceeds 0.3 g/t, whereas the red box marks the highest gold values in the drill hole section. This plot is scaled to the length of the MM106D drill hole. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

due to potassic sericitic alteration adjacent to the veins. In feldspar-quartz porphyry, the inconsistent enrichment in both CO_2 and K_2O in the alteration halos of the veins, the pervasive sericite and carbonate alteration across the porphyry, and the poor correlation between Au and alteration indices, reflect the emplacement of the gold-bearing veins within a much broader sericite-carbonate alteration envelope that extends for as much as 250 m beyond the mineralized zones. This extensive sericite-carbonate alteration is consistent with the lack of unaltered protoliths near the gold deposit. The host rocks at the Greenstone deposit comprise quartz-plagioclase-sericite-chlorite $\pm$ biotite mineral assemblages that are consistent with mid- to upper greenschist facies metamorphic conditions. Hydrothermal alteration selvages along vein walls contain higher proportions of sericite and chlorite.

Pathfinder metals with high Au Spearman Rank correlation coefficients (S, W, Te, Mo, Bi, Se, and As in sandstone; W, As, Te, and S in porphyry) generally correspond with major mass gains of those elements in isocon diagrams and % mass change histograms of V_1 and V_3 veins. Enrichments in Au, Ag, As, Bi, Co, Ni, Pb, Sb, and Te are also present in inclusion-rich pyrite on trace element maps. Point concentrations of Au, Bi, Cr, Pb, Te, Sb, Ti, and As within zones of more uniform enrichments of those elements suggest that they represent inclusions in pyrite grains, which also incorporated those metals either as submicroscopic particles or within their crystal lattice during their growth. Inclusion-poor pyrite

with primary crystallographic Ni-Co-As zoning contains no or negligible gold, which is consistent with the lack of correlation between Au and Ni-Co in whole rock samples. Elevated Au, Ag, Bi, Pb, Sb, and Te concentrations along late fractures intersecting both pyrite types indicate that Au was either remobilized or additional Au was introduced during an event that postdates both the early- D_1 and the syn- D_2 gold episodes. The strong Au-W correlation in whole rock samples and the low concentrations of W in both inclusion-rich and inclusion-poor pyrite suggest that W was included in other minerals (scheelite(?)) observed by Horwood and Pye, 1955) that formed during gold deposition.

In BIF, sulfide-rich alteration halos surrounding quartz-carbonate veins yielded gold grades as high as 89.9 g/t (Tóth et al., 2023b). The alteration halos formed by replacement of magnetite-rich laminae by pyrite and arsenopyrite ($\pm$ chalcopyrite, pyrrhotite). Sulfidation reactions between magnetite/ilmenite and the hydrothermal fluids destabilized Au-bisulfide complexes in the fluids, precipitating gold (Groves et al., 1998, 2003; Phillips and Powell, 2010) and rutile as a by-product of the sulfidation reactions (Clark and Williams-Jones, 2004). In feldspar-quartz porphyry and sandstone, the replacement of titanomagnetite and/or ilmenite by pyrite and rutile (Fig. 10G), and the presence of rutile inclusions in pyrite (Fig. 10E), further suggest that host rock sulfidation was one of the processes by which gold precipitated. Drops in fluid pressure during the formation of extensional veins

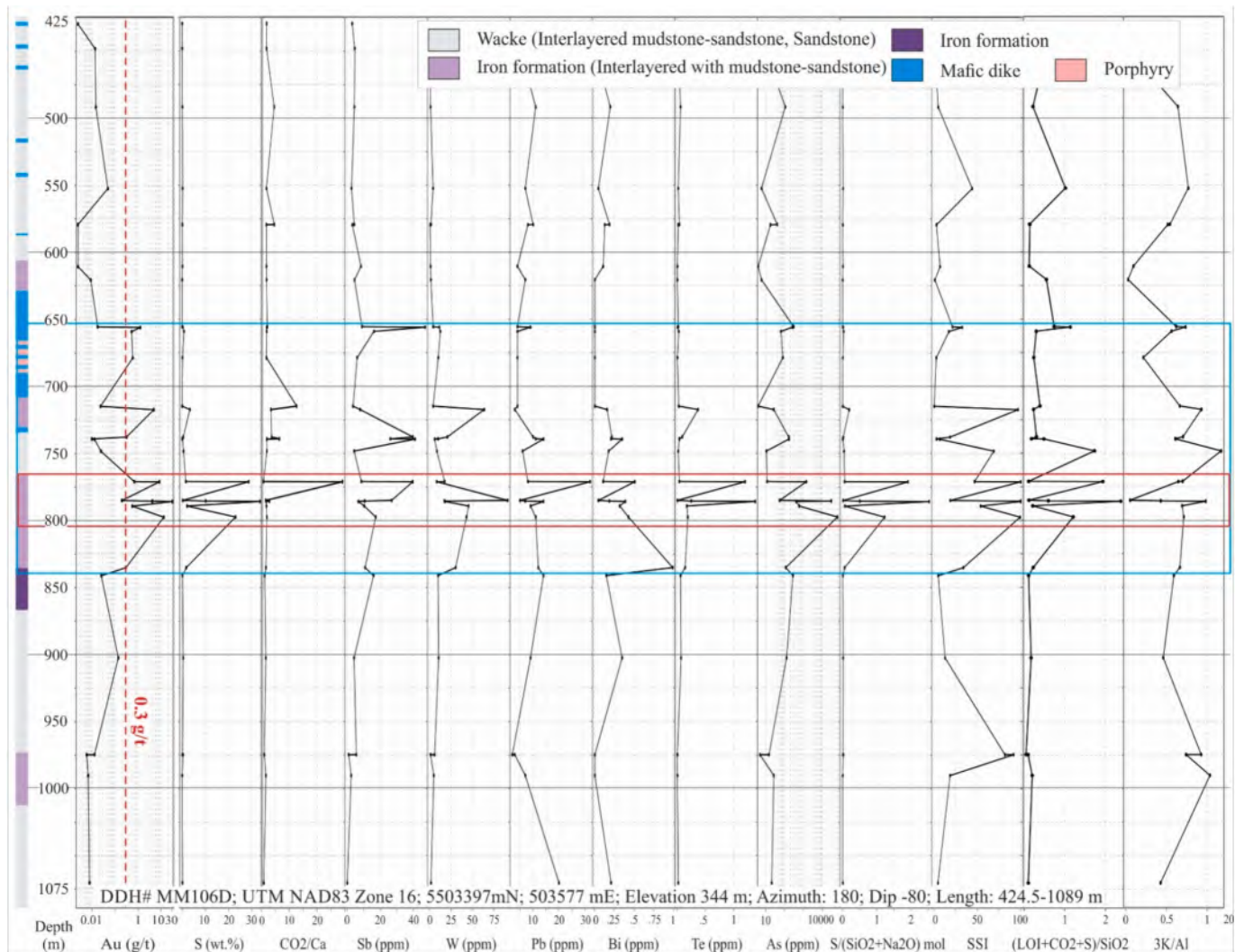


Fig. 14. Geochemical profile of pathfinder elements and alteration indices in drill hole # MM106D across section 503575E, UTM NAD 83, Zone 16. Fig. 2 shows the location of the drill hole. The blue box outlines the interval where Au exceeds 0.3 g/t, whereas the red box marks the highest gold values in the drill hole section. This plot is scaled to the length of the MM106D drill hole. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

further precipitated gold in the veins (Figs. 8B and 8E). Carbonation and fluid mixing may also contribute to gold precipitation as these processes were invoked to explain the isotopic (S, O, H) signatures of gold-associated mineral phases in Fe-poor host rocks (e.g. [Palin and Xu, 2000](#); [Bateman and Hagemann, 2004](#)).

The presence of auriferous inclusion-rich pyrite and barren inclusion-poor pyrite indicates that they grew from pulses of metal-rich and metal-poor fluids. The latter fluid may have evolved from the metal-rich fluid through continuous fluid-rock reaction. Gold and other metals present in the porous pyrite were subsequently redistributed or was deposited from new pulses of gold-rich fluids, as suggested by the filling of fractures in pyrite and quartz-carbonate veins by native gold (Fig. 10F), Au-Ag-tellurides and galena, and by the presence of Ag-Bi ± Pb-telluride and native gold in rutile inclusions in pyrite (Fig. 7D-E, Fig. 9C). The deposition of late gold is further supported by Au-rich fractures that crosscut barren inclusion-poor pyrite (Fig. 16). This late gold deposition or remobilization along fractures and inclusions post-date both the early-D₁ and syn-D₂ mineralization. We hypothesize that the presence of subhedral to euhedral pyrite and arsenopyrite as well as pyrrhotite may indicate an increase in metamorphic grade. Telluride minerals lower the melting point of sulfide-rich ore deposits to the lowest greenschist to amphibolite facies metamorphic conditions

([Tomkins et al., 2007](#)) resulting in the liberation of gold and low-melting point chalcophile elements during the recrystallization of pyrite, and precipitation of gold and tellurides along fractures and inclusions during prograde metamorphism of the Quetico subprovince to the south.

In summary, the similarity in the mineralogy and mass changes in alteration halos of V₁ and V₃ quartz-carbonate veins suggest that they were deposited from hydrothermal fluids similar in composition. The deposition of tourmaline-quartz V₂ veins indicates a transient influx of more boron-rich hydrothermal fluids. Textural relationships reveal complex hydrothermal alteration histories involving early stages of sulfide minerals, and possibly gold deposition, during sulfidation reactions between iron-rich oxide minerals and hydrothermal fluids that infiltrated the wall rocks of the veins. Metal-rich and metal-poor pyrite reflect the metal content of the reacting hydrothermal fluids. Auriferous pyrite deposition was followed by the redistribution and/or introduction of gold and other metals by metal-rich late fluids which percolated along fractures in earlier-formed pyrite and precipitated gold and other metal-rich mineral phases along these fractures. Hydrothermal alteration produced a broad sericite-carbonate footprint, which extends as far as 250 m from the deposits, in which S, Te, As, W, and Bi are the best indicators of the presence of Au.

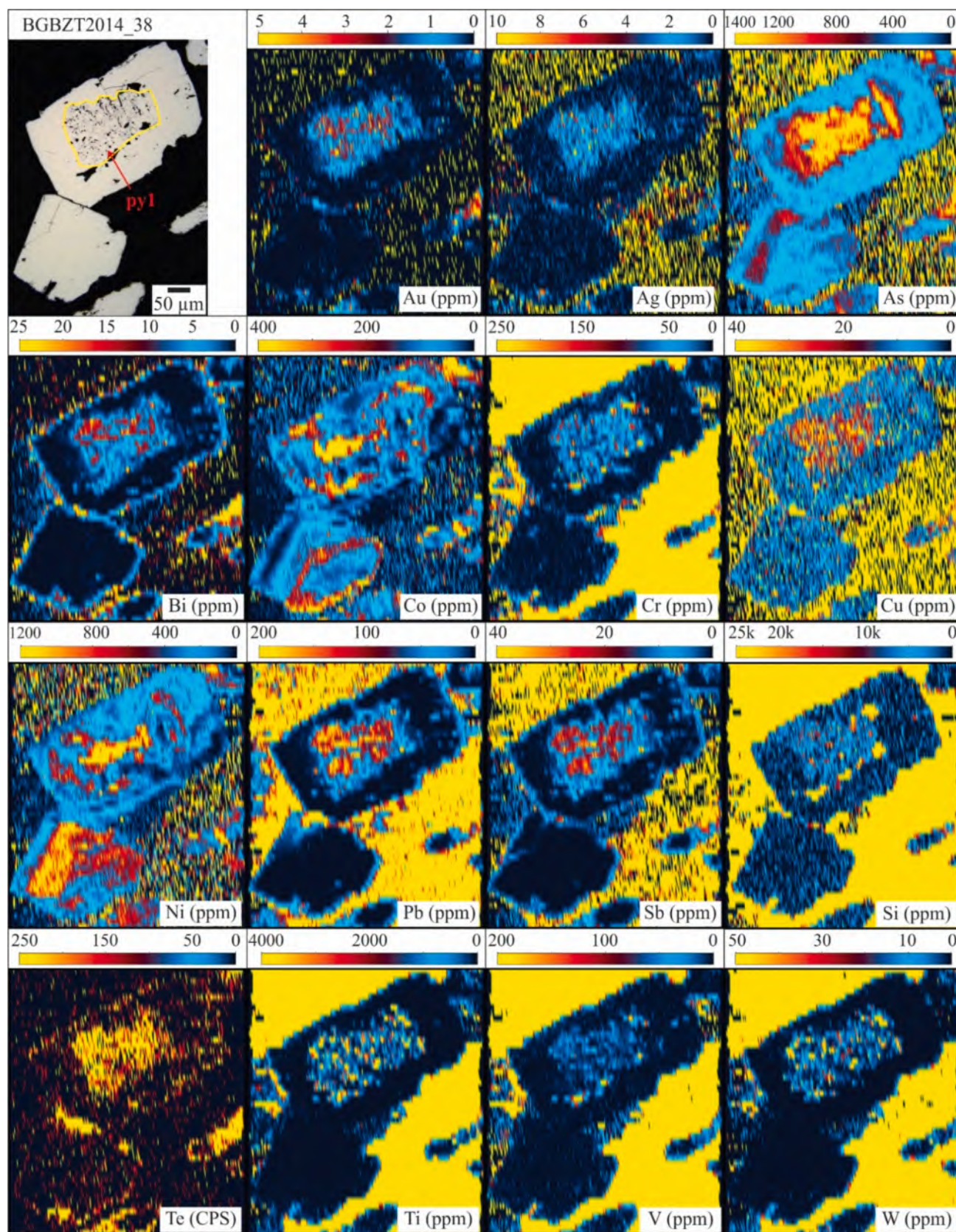


Fig. 15. LA-ICP-MS elemental map of pyrite associated with quartz-carbonate V_1 vein in sandstone. Reflected light photomicrograph at upper left of diagram showing an inclusion-poor pyrite with an inclusion-rich core (py1) and a second lower inclusion-poor pyrite. See Supplementary Material 1 for the Fe-concentration (count per second (CPS)) map.

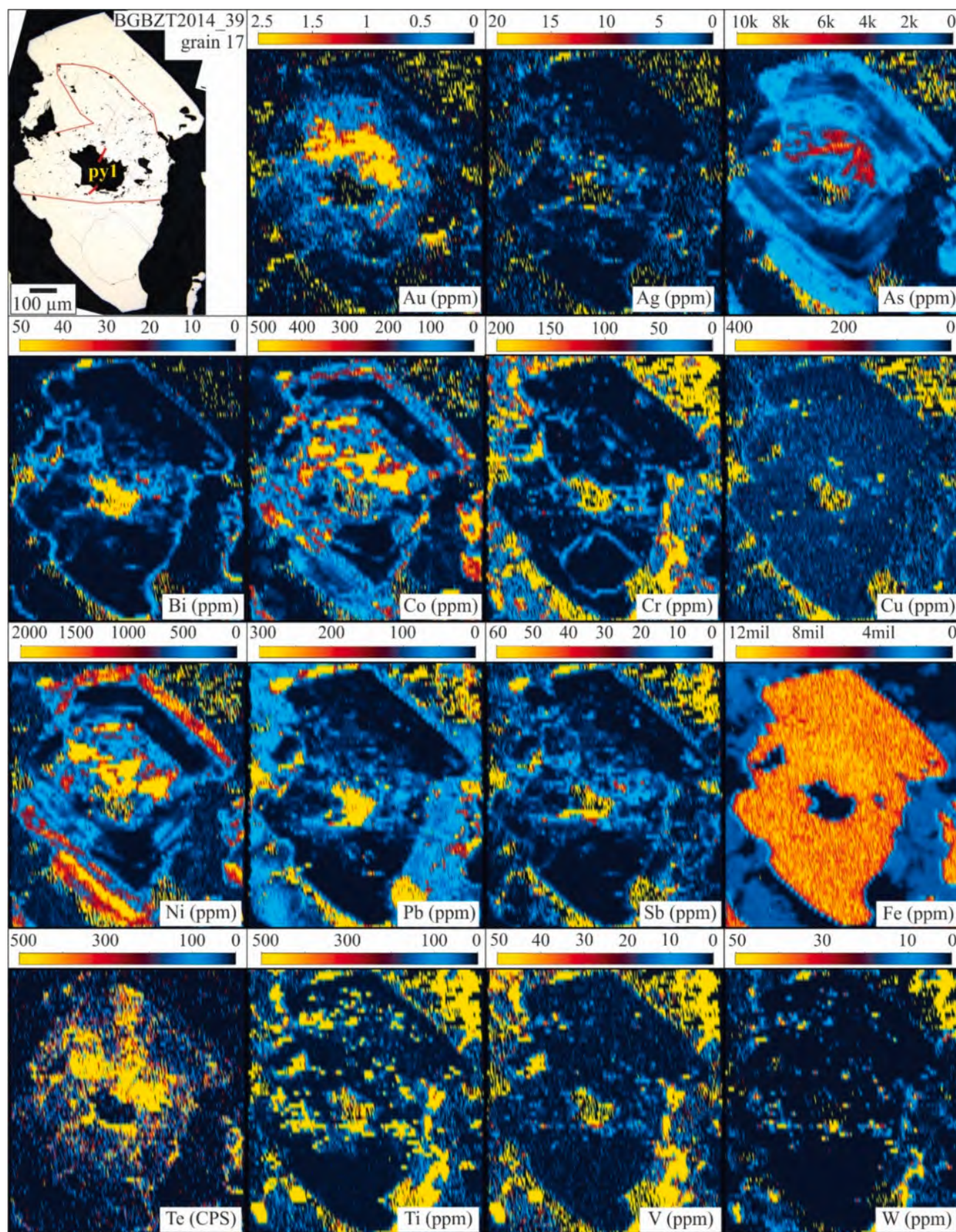


Fig. 16. LA-ICP-MS elemental map of pyrite associated with quartz-carbonate V_3 vein-related gold mineralization in sandstone. Reflected light photomicrograph at upper left of diagram showing an inclusion-poor pyrite with an inclusion-rich core (py1).

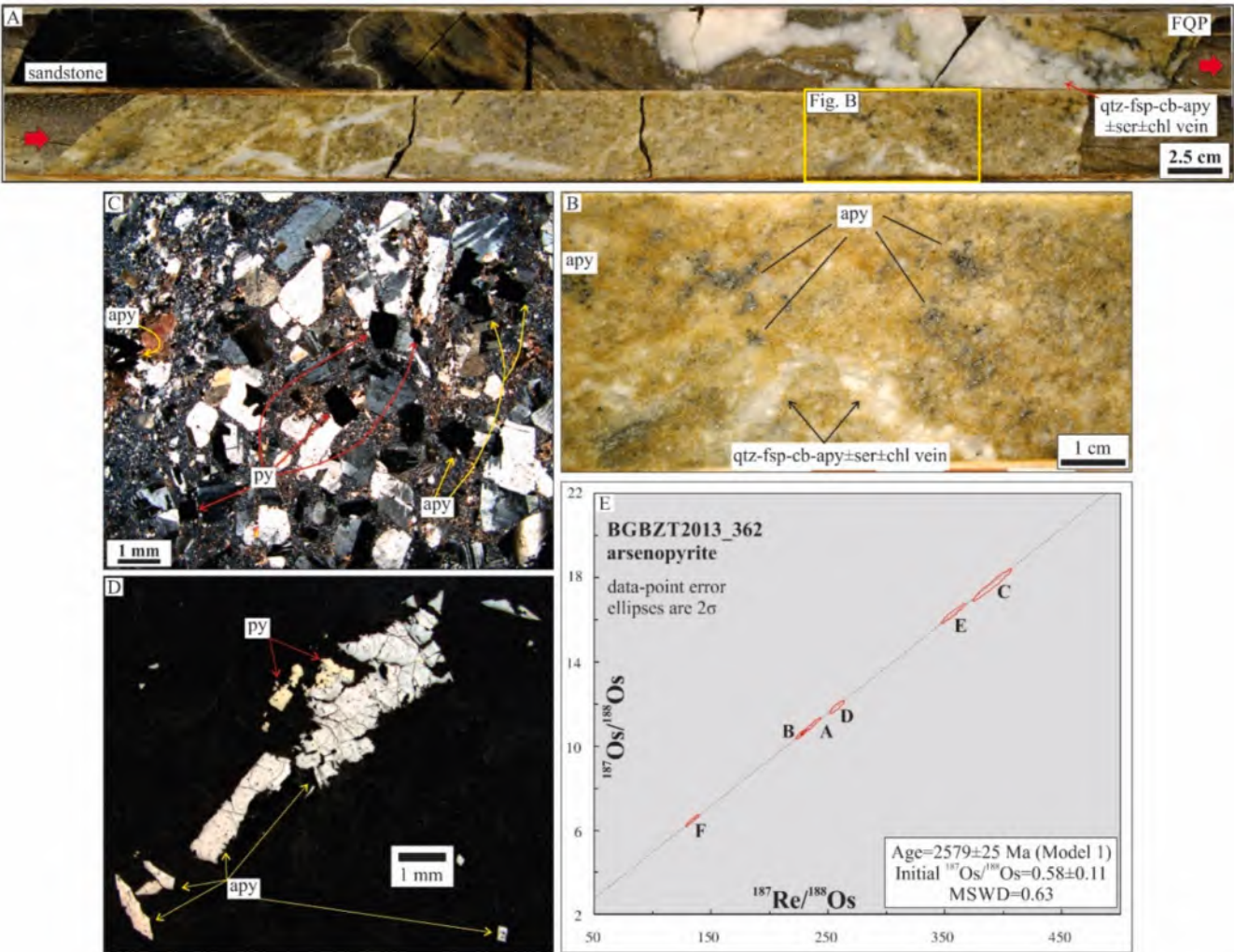


Fig. 17. A-B) Drill core photograph of arsenopyrite-rich gold mineralization associated with quartz-carbonate veins adjacent to contact between feldspar-quartz porphyry and sandstone. The yellow box marks the location of Fig. 17B. qtz: quartz; fsp: feldspar; cb: carbonate; apy: arsenopyrite; ser: sericite; chl: chlorite; C) Cross-polarized transmitted light photomicrograph of arsenopyrite (apy) and pyrite (py) in feldspar-quartz porphyry; D) Reflected light microphotograph of arsenopyrite and pyrite in the feldspar-quartz porphyry. E) Re-Os isochron diagram for arsenopyrite in feldspar-quartz porphyry. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Re-Os isotope data of arsenopyrite in mineralized zone from sample BGBZT2013_362 (DDH: MM276; depth: 305.4 m to 306.49 m).

Sample ID	Re (ppb)	± 2σ	Os (ppt)	± 2σ	$^{187}\text{Re}/^{188}\text{Os}$	± 2σ	$^{187}\text{Os}/^{188}\text{Os}$	± 2σ	Model age (Ma)	± 2σ
BGBZT-2013-362-A	0.94	0	46.3	1.2	235.5	7.1	10.94	0.33	2603	19
BGBZT-2013-362-B	0.938	0	47	0.6	227.1	3.6	10.55	0.18	2600	22
BGBZT-2013-362-C	0.693	0	28.1	1	390.4	13.9	17.65	0.63	2580	21
BGBZT-2013-362-D	0.852	0	40.3	0.7	257.5	5.2	11.85	0.26	2588	26
BGBZT-2013-362-E	0.93	0	39	0.9	356.9	8.6	16.28	0.39	2597	16
BGBZT-2013-362-F	1.121	0.01	73.5	1.9	134.1	5	6.46	0.24	2608	29

4.3. Chronology of gold mineralization events

The 2579 ± 25 Ma (2σ) Re-Os arsenopyrite model age for gold mineralization at the Greenstone deposit is significantly younger than that of other mineralization events in gold camps across the Superior Province. The arsenopyrite grains may have precipitated from late hydrothermal fluid pulses generated during regional metamorphism of the Quetico metasedimentary subprovince. According to Percival and Sullivan (1988), regional metamorphism and large-scale plutonism occurred between 2670 and 2650 Ma in the Quetico subprovince; however, fluid circulation may have lasted much longer due to

protracted metamorphism and slow cooling rates during the late Archean (Chowdhury et al., 2021).

In the Red Lake gold camp in the Uchi subprovince, the main stage of gold mineralization occurred synchronously with D₂ regional deformation at ca. 2723 Ma to ca. 2712 Ma (Dubé et al., 2004). In the Timmins camp in the Abitibi subprovince, gold mineralization occurred later and is syn- to late-main phase shortening (D₃) between circa 2660 Ma and 2640 ± 10 Ma (Dubé et al., 2020; Herzog et al., 2022). Circa 100 km to the southeast in the Kirkland Lake – Larder Lake camp area, gold was deposited between 2672 ± 2 Ma and 2665 ± 4 Ma during D₂ regional folding and shearing (Wilkinson et al., 1999), but was later remobilized

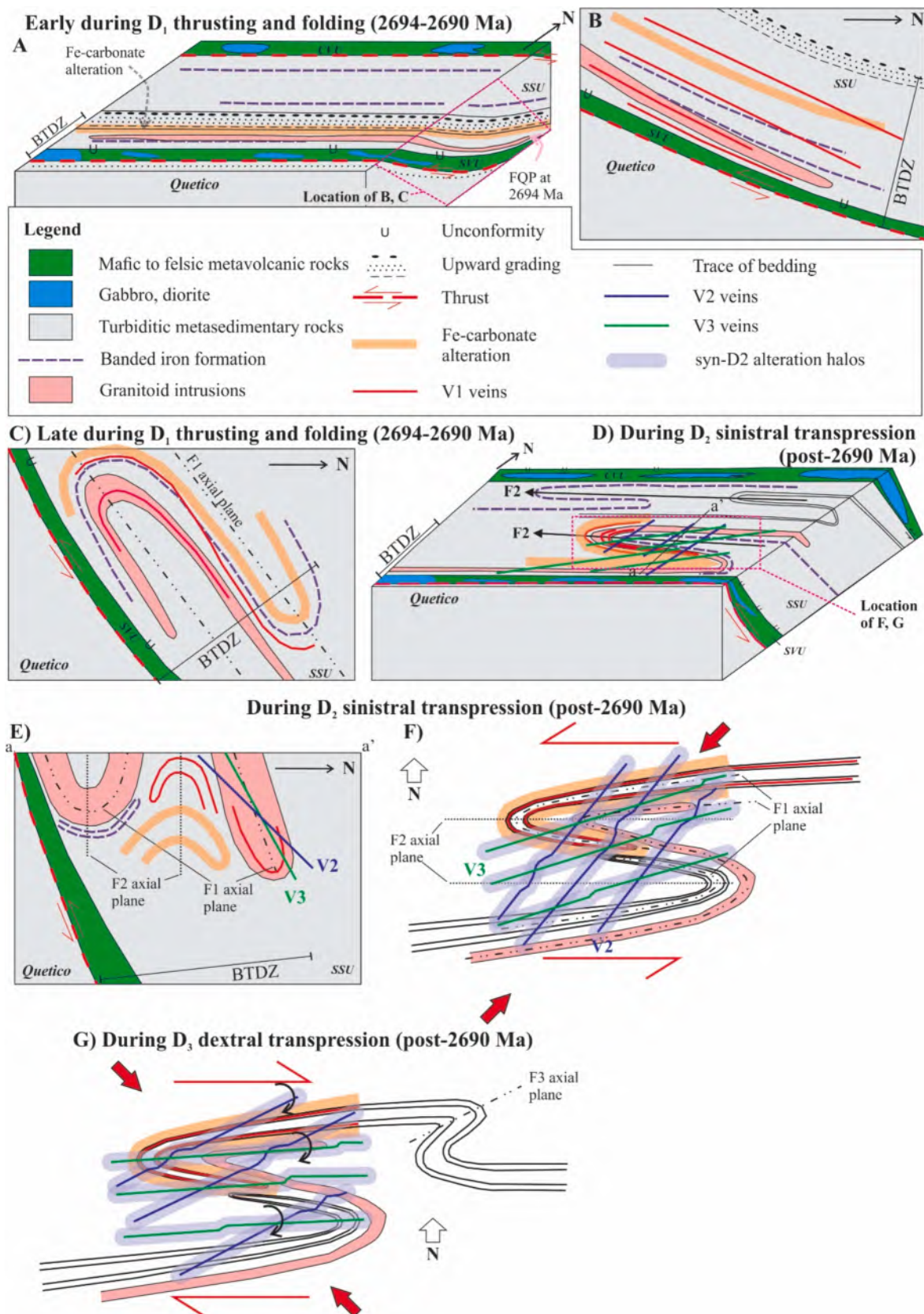


Fig. 18. Schematic model for the emplacement of V₁ to V₃ veins: A) 3D block model and B) cross-section of the emplacement of early bedding-parallel Fe-carbonate alteration and quartz-carbonate V₁ veins during D₁ thrusting and folding event; C) Folding of the early bedding-parallel Fe-carbonate alteration and quartz-carbonate V₁ veins later during the D₁ event; D) Block diagram, E) cross-section F) map view of syn-D₂ events: Overprinting of F₁ folds by S-shaped F₂ folds during D₂ sinistral transposition; Emplacement of tourmaline-quartz V₂ veins, rotation of the veins, and emplacement of quartz-carbonate V₃ veins later during the D₂ event; G) Map view of syn-D₃ events: Rotation and stretching V₂ and V₃ veins and formation of F₃ flanking structures adjacent to the veins during D₃ dextral transposition.

during the Paleoproterozoic in hematite-quartz-carbonate veins (1730 ± 5 Ma, U-Pb monazite), which overprint the main mineralized veins at the Young-Davidson deposit (Zhang et al., 2014). Closer to the BGB, gold at the Hemlo deposit in the Wawa subprovince was deposited and/or remobilized during a short, contracted, regional D₂ deformation, magmatic and mineralization event (Lin, 2001; Muir, 2002, 2003), which is bracketed between 2680 ± 1 and 2677 ± 1 Ma (Davis and Lin, 2003). More recently, Poulsen et al. (2020) interpreted the Hemlo deposit as a deformed and metamorphosed gold system formed from oxidized hydrothermal fluids in an upper crustal setting with the main mineralization phase between 2686 Ma and 2682 Ma.

In other parts of the Wabigoon subprovince, the Goldlund deposit formed during sinistral transpression along the NE-trending Manitou Dinorwic deformation zone during two events dated at ca. 2672 to ca. 2626 Ma and ca. 2590 Ma to ca. 2580 Ma (U-Pb xenotime; Zammit et al., 2022). Gold mineralization at the Hammond Reef deposit, which is located 200 km WSW of the BGB and 120 km SSE of the Goldlund deposit, is however significantly older at ca. 2700 Ma to 2690 Ma, although it was emplaced within a similar NE-trending sinistral deformation zone (Fouillard, 2023). This age range is consistent with the deposition of gold in the Geraldton camp, which has a maximum age of 2694 ± 1 Ma obtained from pre-or early-D₁ altered porphyry at the Greenstone deposit (Tóth et al., 2022). The gold mineralization is deposited in second-order structures adjacent to the crustal-scale Bankfield-Tombill fault that was initiated and reactivated during deformation events related to the ca. 2690 Ma Shebandovian orogeny, a major tectonic process responsible for the assembly of the Superior craton (Percival et al., 2012). The 2579 ± 25 Ma Re-Os model age is broadly similar to Pb model ages (2474–2563 Ma) obtained from galena collected at the Bankfield mine (#1 on Fig. 1; Anglin, 1987) and the U-Pb xenotime age of ca. 2590 Ma to ca. 2580 Ma at the Goldlund deposit (Zammit et al., 2022). As these ages are significantly younger than those of other major mineralization events in the Superior craton and much postdate the Shebandovian orogeny, they likely represent new pulses of hydrothermal fluids that postdate the main mineralization events at these deposits.

4.4. Comparison with other major orogenic gold camps in the Superior Province

The superimposition and recurrence of mineralization events associated with major faults or shear zones occur in other Archean orogenic gold camps. In the Timmins camp, ca. 2672 ± 7 Ma (Ayer et al., 2005) Cu-Au-Ag-Mo stockwork mineralization is spatially associated with the 2691 ± 3 Ma Millerton and 2689 ± 1 Ma Pearl Lake porphyries (Corfu et al., 1989), which are located within ca. 5 km north of the Porcupine – Destor deformation zone (Burrows et al., 1993; Dubé et al., 2018). This early style of mineralization is overprinted by gold-bearing quartz-carbonate veins that were emplaced during D₃ sinistral and D₄ dextral transpression along the Hollinger shear zone, a second-order structure parallel to the Porcupine – Destor deformation zone; however, the bulk of the gold mineralization was deposited syn- to late-D₃ (Davies and Luhta, 1978; Mason and Melnik, 1986; Corfu et al., 1989; Burrows et al., 1993; Bleeker, 1995, 1999; Dubé et al., 2003; Ayer et al., 2005; Bateman et al., 2005, 2008; Dubé et al., 2018; Dubé et al., 2020). In the Kirkland Lake – Larder Lake camp, orogenic gold mineralization was emplaced during D₂ south-side-up shearing along the Larder Lake – Cadillac deformation zone (Ayer et al., 2005; Ispolatov et al., 2008; Lafrance, 2015) and during D₄ reverse-dextral displacement along the Kirkland Lake fault (Ispolatov et al., 2008), which is one of several narrow brittle-ductile faults located 2 – 3 km north of the Larder Lake – Cadillac deformation zone. The tectonic evolution of both the Porcupine – Destor and the Larder Lake – Cadillac deformation zones and the gold events in the Timmins and Kirkland Lake camps are associated with a craton-scale contraction caused by the amalgamation of the Wawa-Abitibi and Minnesota River Valley terranes and the closure of the Pontiac basin

(Bateman et al., 2008; Percival et al., 2012; Dubé et al., 2018).

In summary, as with other major orogenic gold camps in the Superior Province, gold deposits in the Geraldton camp formed during multiple mineralization events along a reactivated major deformation zone or fault associated with contractional deformation during the assembly of the Superior craton. Gold was deposited from syntectonic gold-bearing fluid pulses which contributed to and culminated in the formation of the Geraldton gold camp.

5. Conclusions

Previous interpretations invoked orogen-parallel dextral transcurrent faulting as the main structural control on the emplacement of gold mineralization in the BGB (Anglin, 1987; Macdonald, 1988; Williams, 1989; Lafrance et al., 2004; DeWolfe et al., 2007; Lavigne, 2009). Gold mineralization was thought to be coeval with the formation of structures and dilational zones during regional dextral transcurrent faulting, such as the hinge of Z-shaped folds, dextral shear zones at the contact between rock units, and reactivated thrust faults at the contact between panels of metasedimentary and metavolcanic rocks. Our study indicates that gold was emplaced relatively earlier during the structural history of the belt and involved two stages of gold mineralization. The hydrothermal mineralization history of the Geraldton camp began with the formation of quartz-carbonate veins and early bedding-parallel Fe-carbonate alteration during D₁ faulting and thrust imbrication of the belt. These veins and early Fe-carbonate alteration were folded and transposed along the Bankfield-Tombill deformation zone as panels in the belt were regionally folded and shear became localized along higher strain deformation zone during regional D₂ sinistral transpression. Hydrothermal fluids flowed into the Bankfield-Tombill deformation zone which acted as a conduit for the upward migration of hydrothermal fluids, and auriferous tourmaline-quartz and quartz-carbonate veins were deposited across S-shaped folds that formed during sinistral shearing along the deformation zone. Reactivation of the deformation zone during regional D₃ dextral transpression rotated and boudinaged V₂ and V₃ veins and redistributed or introduced new gold and other metals into localized dextral faults and shear zones.

The complex hydrothermal fluid history of the Geraldton camp is recorded by the textures of pyrite grains and their chemical composition. Inclusion-poor pyrite and metal-rich inclusion-rich pyrite are associated with both *syn*-D₁ and *syn*-D₂ vein generations. They formed during fluid-rock sulfidation reactions with their metal content reflecting that of the hydrothermal fluids. Gold was then redistributed, or new gold and other metals were introduced, during the infiltration of late fluids along microfractures cutting across sulfide minerals and other alteration minerals. As this process occurred at different times during the formation of the Greenstone deposit, our study exemplifies how the juxtaposition of multiple mineralization events and the redistribution of metals during the prolonged structural history of a major deformation zone can result in the formation of complex ore deposits.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zsuzsanna Toth reports financial support, equipment, drugs, or supplies, and travel were provided by Greenstone Gold Mines. Zsuzsanna Toth reports financial support, equipment, drugs, or supplies, and travel were provided by Premier Gold Mines Limited. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Appendix A. Analytical methods

Petrography

Over 200 polished thin sections were prepared of least altered to strongly altered mineralized samples. Semi-quantitative mineral chemical analyses were collected using an Oxford energy-dispersive spectrometer mounted on the JEOL JSM-6400 scanning electron microscope (SEM) at the Central Analytical Facilities of Laurentian University, using 20 kV acceleration voltage and a 1.005 nA beam current at 15 mm working distance.

Mineral-chemical mapping of sulfide minerals

The mineral chemistry of gold-prospective pyrite and arsenopyrite grains were analyzed in situ on polished thin sections using the Thermo X Series II inductively coupled plasma mass spectrometer (ICP-MS) interfaced with a Resonetics RESOLUTION M-50 193 nm ArF excimer laser at the Chemical Fingerprinting Laboratory, Laurentian University, Sudbury, ON. The beam diameters were chosen based on the size of the grain of interest and the type of analyses to be carried out. For elemental mapping the beam diameter varied between 14 and 26 μm . The scan rates ranged between 7 and 13 $\mu\text{m/s}$, i.e., 1/3 to 1/2 of the beam diameter per second. The laser pulse rate, the fluence, and the resulting energy density was 8 Hz, 60 mJ, and 5 J/cm², respectively. Ultra-pure He flowing at a constant rate of 650 ml/min was used as transport gas and 6 ml/min N₂ was added to the cell flow to increase sensitivity (Durrant, 1994).

For the quantification of the trace element concentrations of the analyzed sulfide minerals, the stoichiometric Fe composition was used as the internal standard and calibrations were done using certified reference materials Po725 (Au and Pd), GSD or NIST610 (Ag, As, Bi, Co, Cr, Cu, In, Mn, Mo, Ni, Pb, Sb, Se, Si, Sn, Ti, Tl, V, W, Zn, and Te). The data reduction was carried out using Iolite software package (Paton et al., 2011) and the laser mapping add-on packages developed by Petrus et al. (2017).

Whole rock geochemistry

Samples collected in 2012, 2013, and 2016 were analyzed at Activation Laboratories Ltd. in Ancaster, Ontario (year of collection indicated in the sample ID). The samples were crushed in a Cr-free steel mill and dissolved by metaborate/tetraborate, 4-acid, Aqua Regia or sodium peroxide digestion prior to whole rock geochemical analyses using fusion X-ray fluorescence spectrometry (XRF) and inductively coupled plasma mass spectrometry (ICP-MS) to obtain major oxide and trace element compositions, respectively. The analytical methods are presented along with the composition of representative samples in Supplementary Material 1 whereas the entire dataset is given in Tóth et al. (2023b). Gold values were determined by atomic absorption or gravimetric fire assay methods when Au < 3000 ppb and Au > 3000 ppb, respectively. The latter method was also used for ore grade Ag analyses. Total S, CO₂, and FeO concentrations were determined by infrared

absorption and titration, respectively. Arsenic values were analyzed by ICP-MS following peroxide fusion if their concentration exceeded 10,000 ppm.

The samples collected in 2014 were crushed by agate mill and dissolved using closed vessel multi-acid or Aqua Regia digestion at the Geoscience Laboratories of the Ministry of Northern Developments and Mines in Sudbury, Ontario. Major oxide and trace elements were analyzed using fusion X-ray fluorescent spectrometry (XRF) and inductively coupled plasma mass spectrometry (ICP-MS), respectively. Both analytical accuracy and precision are commonly good to excellent (Percent relative difference {%RD} ≤ 10 and percent relative standard deviation {%RSD} ≤ 10); however, both accuracy and precision deteriorate below the limit of quantification (LOQ) and approaching the limit of detection (LOD; LOQ $\sim 3.3 \times \text{LOD}$) (Jenner, 1996; Piercey, 2014 and references therein). Additionally, the gold analyses perform more poorly at precision and accuracy tests due to the nugget effect observed in orogenic gold deposits.

Re-Os geochronology

Re-Os isotope dating of arsenopyrite was carried out at the Canadian Centre of Isotopic Microanalysis (CCIM) at the University of Alberta, Canada. The arsenopyrite concentrate derived from the selected sample (BGBZT2013_362C) was first analyzed using isotope dilution negative thermal ionization mass spectrometry (ID-NTIMS) to determine if it had sufficient Re to proceed with the datation. The mineral concentrate was prepared by crushing using metal-free mortar and arsenopyrite fragments were separated using gravity and magnetic methods. Aliquot of the concentrate were placed in a borosilicate glass Carius tube with a known amount of a mixed ¹⁸⁵Re + ¹⁹⁰Os spike, and digested at 220°C for 48 h in inverse aqua regia. Osmium and rhenium were chemically separated and purified as described in Shirey and Walker (1995), Cohen and Waters (1996), Birck et al. (1997). The Re and Os isotopic compositions were analyzed by ID-NTIMS as described in detail by Hnatyshin et al. (2020). The Re-Os age was calculated and plotted using Isoplot v. 3.0 (Ludwig, 2003) with errors at the 2 σ level using the decay constant of ¹⁸⁷Re isotope from Smoliar et al. (1996).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.oregeorev.2024.106345>.

Data availability

The whole rock geochemical data has been released in Tóth et al. (2023b). Any additional data will be made available upon request.

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